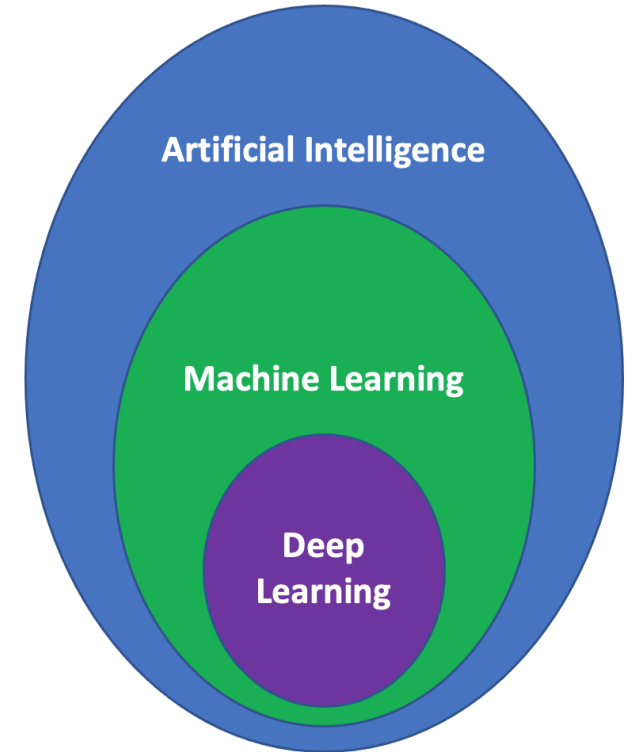


Deep Learning

Lecture 1

What is Deep Learning

- ***Artificial Intelligence (AI)*** any technique that enables computers to mimic human intelligence (e.g., perception, comprehension, attention, problem solving, planning) from multiple modalities
- ***Machine Learning (ML)*** is a subset of AI that learns from data
- ***Deep Learning (DL)*** is a subset of ML that involves many parameters and large amounts of data



First use-case: Image classification

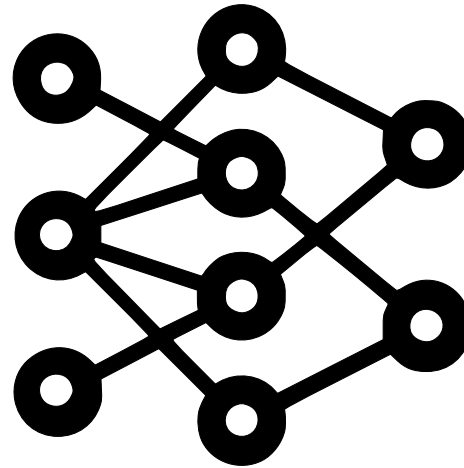
Real world input



Model
input

$\begin{bmatrix} 124 \\ 140 \\ 156 \\ 128 \\ 142 \\ 157 \\ \vdots \end{bmatrix}$

Model



Model
output

$\begin{bmatrix} 0.00 \\ 0.00 \\ 0.01 \\ 0.89 \\ 0.05 \\ 0.00 \\ \vdots \\ 0.01 \end{bmatrix}$

Real world output

Aardvark
Apple
Bee
Bicycle
Bridge
Clown
 $\vdots$

Followed by other tasks: Image segmentation

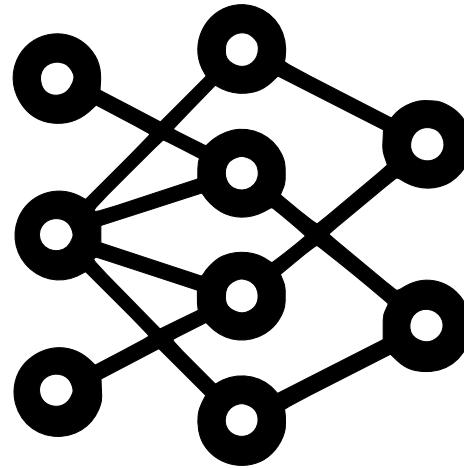
Real world input



Model input

$$\begin{bmatrix} 183 \\ 204 \\ 231 \\ 185 \\ 204 \\ 232 \\ \vdots \end{bmatrix}$$

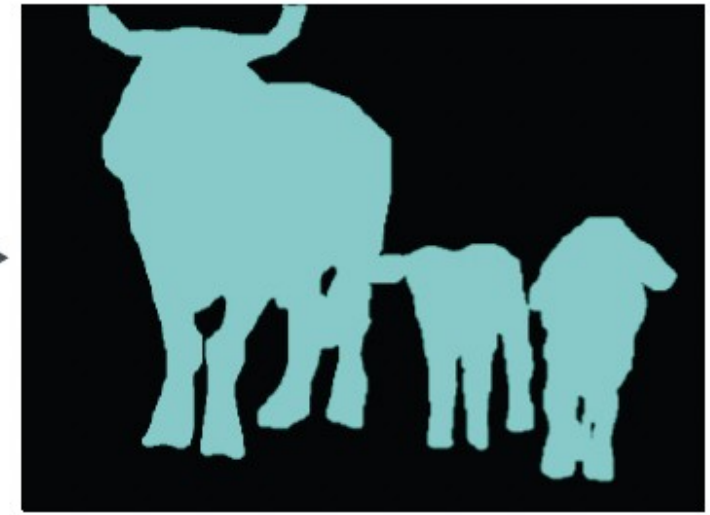
Model



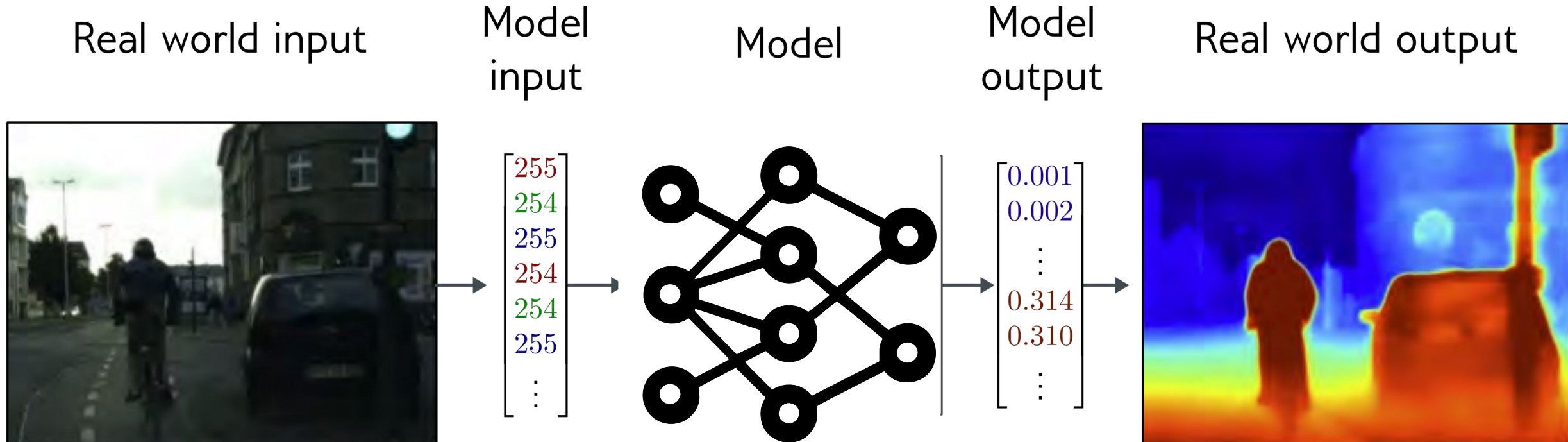
Model output

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \end{bmatrix}$$

Real world output

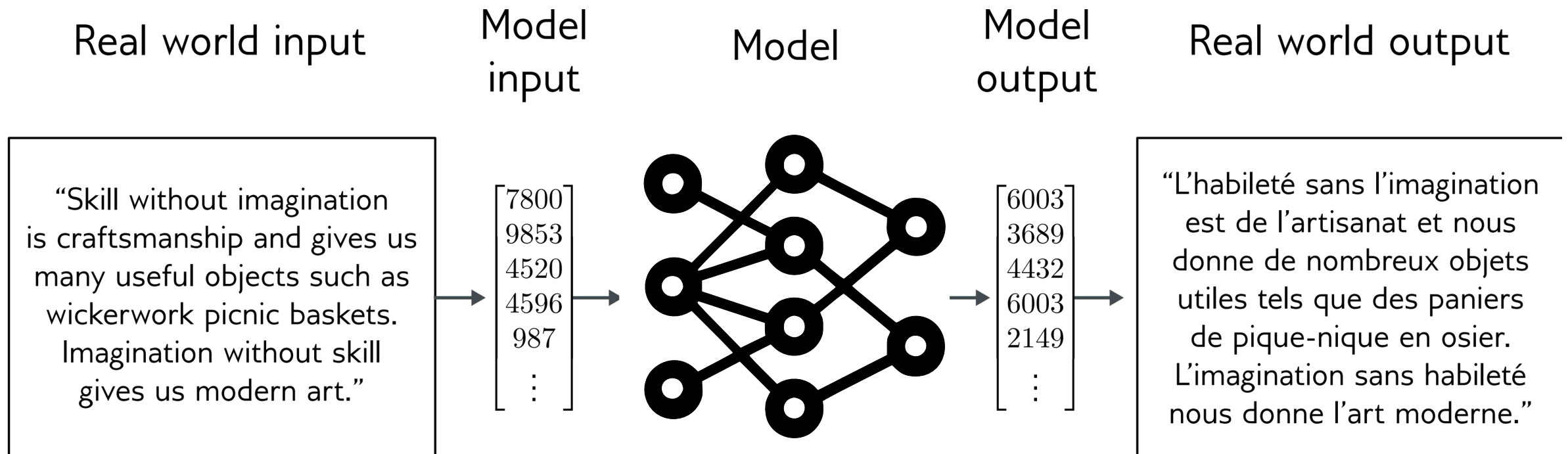


Depth estimation etc



(Lead big rise in autonomous driving cars)

Next, language processing, e.g., Translation



Multimodal: Image captioning

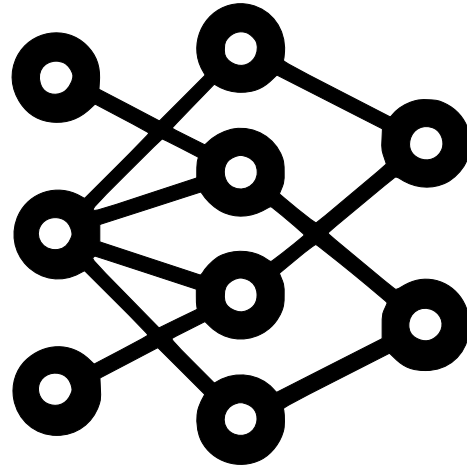
Real world input



Model
input

$$\begin{bmatrix} 183 \\ 204 \\ 231 \\ 185 \\ 204 \\ 232 \\ \vdots \end{bmatrix}$$

Model



Model
output

$$\begin{bmatrix} 1 \\ 5593 \\ 7532 \\ 7924 \\ 1 \\ \vdots \end{bmatrix}$$

Real world output

“A Kazakh man on a
horse holding a
bird of prey”

Followed by Visual Questions

Real world input

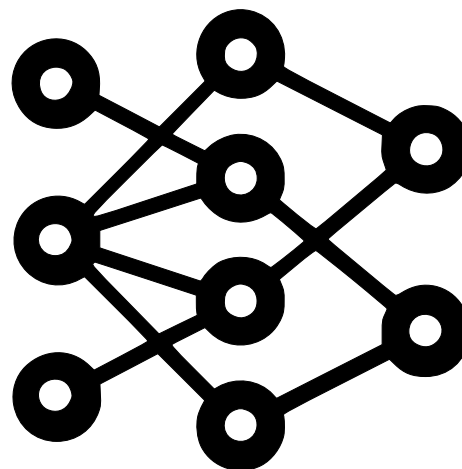


What is unusual?

Model
input

$$\begin{bmatrix} 183 \\ 204 \\ 231 \\ 4596 \\ 987 \\ \vdots \end{bmatrix}$$

Model



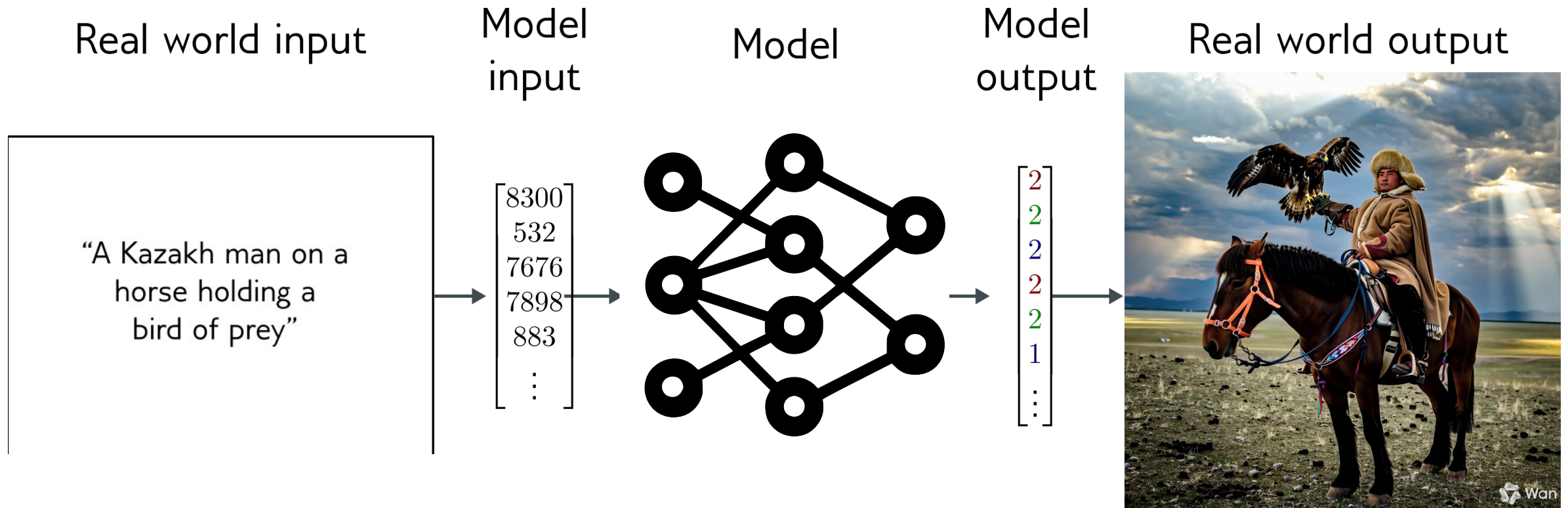
Model
output

$$\begin{bmatrix} 6003 \\ 3689 \\ 4432 \\ 6003 \\ 2149 \\ \vdots \end{bmatrix}$$

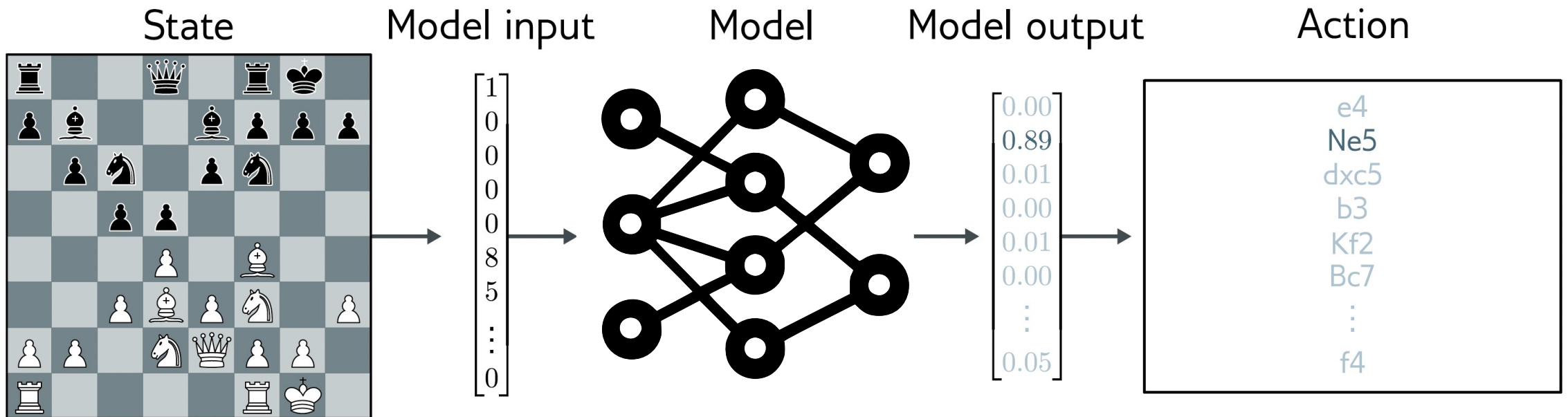
Real world output

The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi

The generative era: Text-to-Image



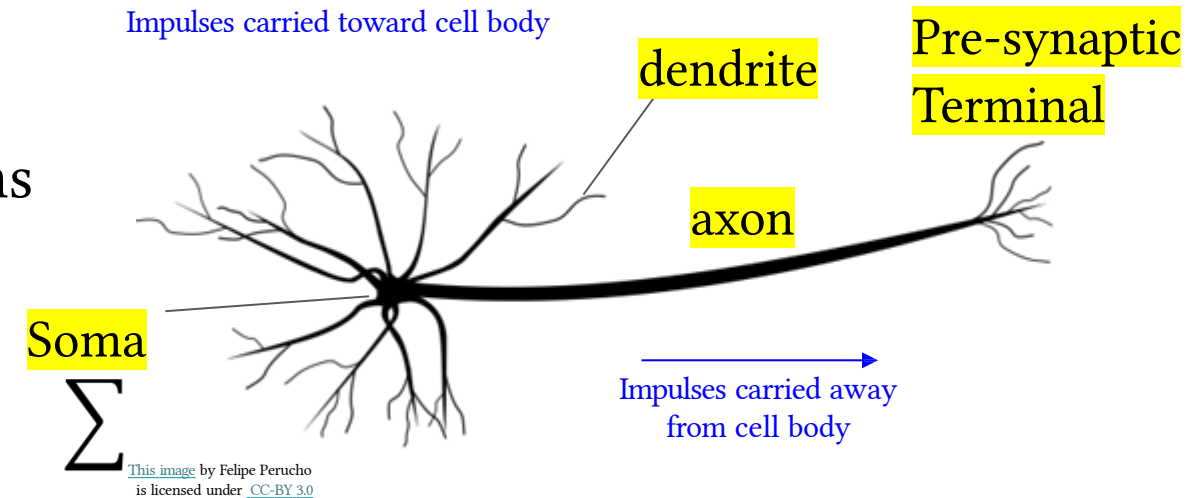
And many other applications, e.g., chess with Deep Reinforcement Learning.



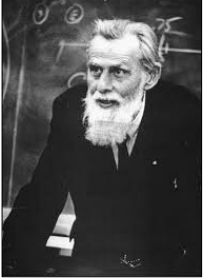
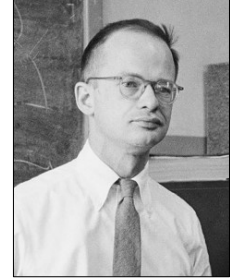
Background: where it started?

The Brain Motivation

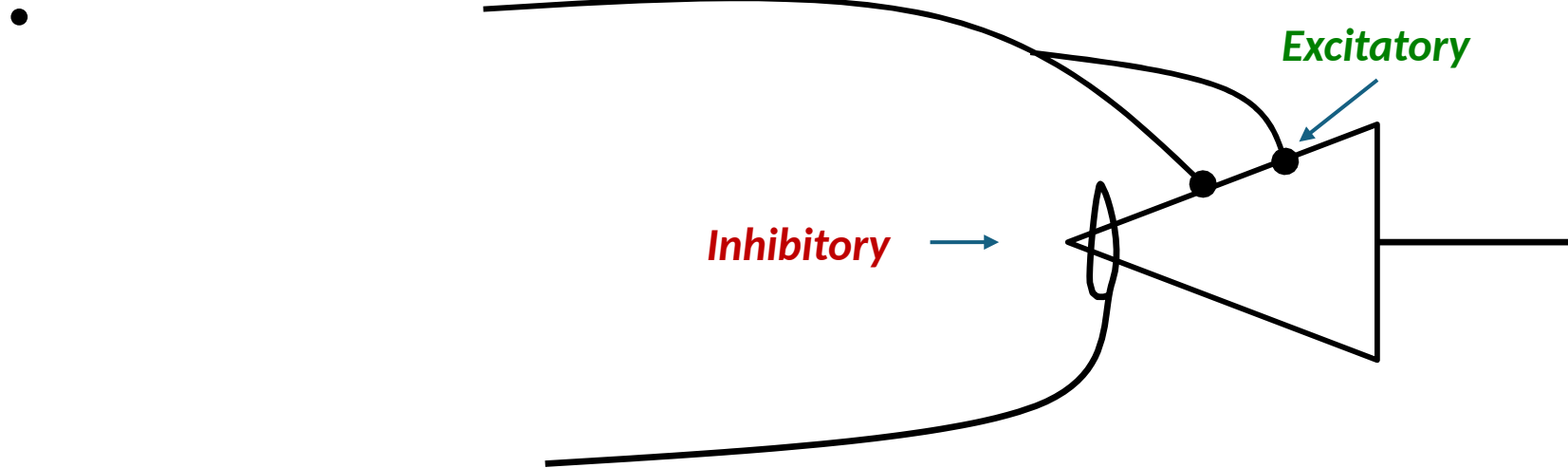
- Late 1800s (Golgi, Cajal): the brain is a network of discrete neurons
- A neuron:
 - Signals enter the neuron via the dendrites into the soma
 - Exit through the axon to other neurons



The McCulloch and Pitts model (1943)



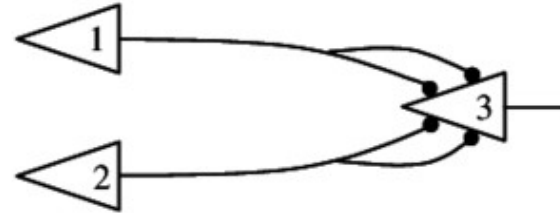
- A mathematical model of a neuron: it fires when the excitatory inputs reach a threshold and no inhibitory input is active



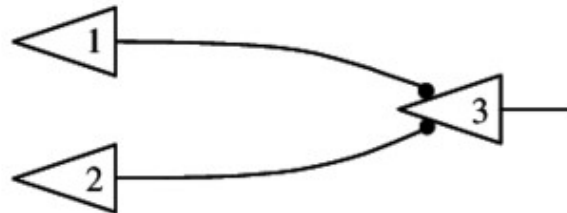
McCulloch and Pitts nets



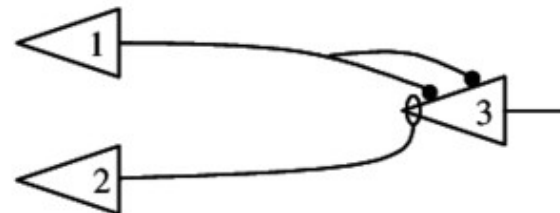
$N_2(t) \Leftrightarrow N_1(t-1)$
net for temporal predecessor



$N_3(t) \Leftrightarrow N_1(t-1) \vee N_2(t-1)$
net for disjunction



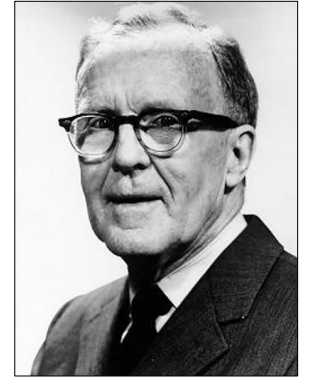
$N_3(t) \Leftrightarrow N_1(t-1) \& N_2(t-1)$
net for conjunction



$N_3(t) \Leftrightarrow N_1(t-1) \& \sim N_2(t-1)$
net for conjunction and negation

However, they didn't provide a learning mechanism

Donald Hebb model (1949)



Learning via correlations:

“When an axon of cell A is near enough to excite a cell B and repeatedly or persistently takes part in firing it, some growth process or metabolic change takes place in one or both cells such that A's efficiency, as one of the cells firing B, is increased.”

(Neurons that fire together wire together)

Hebb's rule as an equation

Notations:

$x \in \mathbb{R}^d$ a feature input vector

$w \in \mathbb{R}^d$ weights for one neuron (this is what we learn)

$y \in \mathbb{R}$ scalar output of the neuron

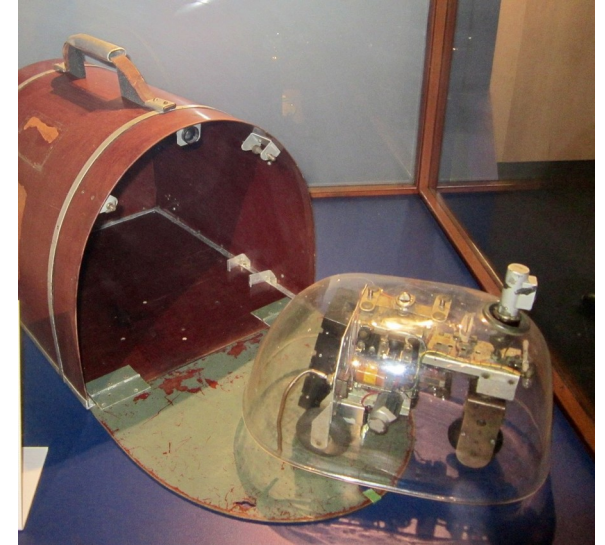
The update rule:

$$y = w^\top x, \quad \Delta w = \eta(xy)$$

Intuition: When x (presynaptic signal) is large **and** y (postsynaptic firing) is large, then Δw is large (η is the learning rate).

Hebb's rule in a machine (1951)

- Grey Walter's CORA (1951) learned associations between two sensors.
- A light sensor caused the robot to move toward light.
- A whistle was repeatedly paired with the light.
- After several pairings, the whistle alone triggered the light-seeking response.
- Simulated on an IBM 704 (1956): the weights grew without bound
- A demonstration, not a product: nothing tells it right from wrong → perceptron (1958)

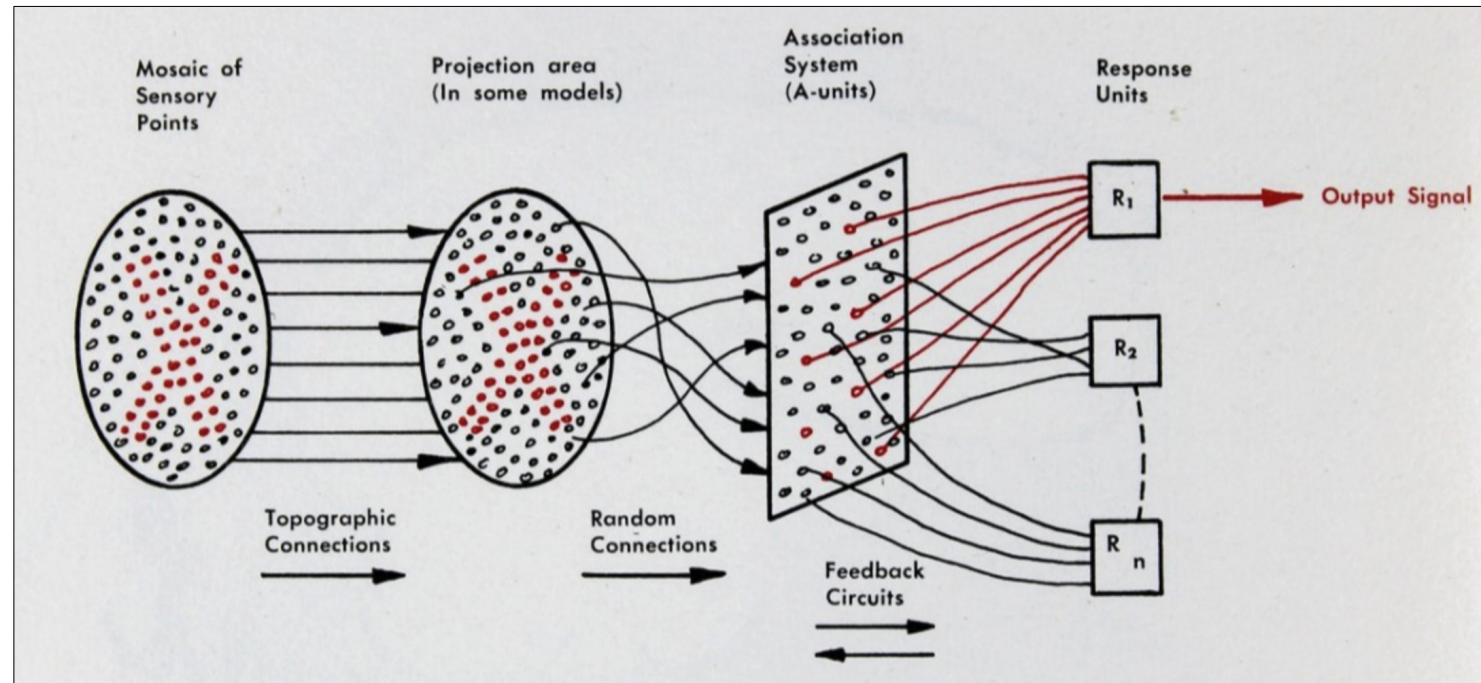
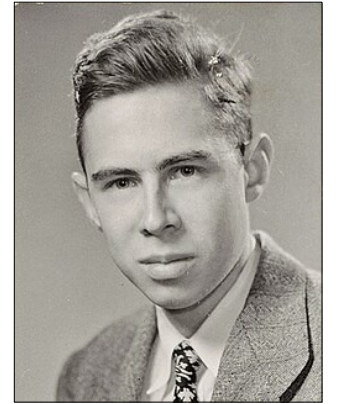


Grey Walter's tortoise, c. 1950. With CORA (1951) it learned to answer a whistle that had been paired with light.

Photo: Ank Kumar, Science Museum London, CC BY-SA 4.0

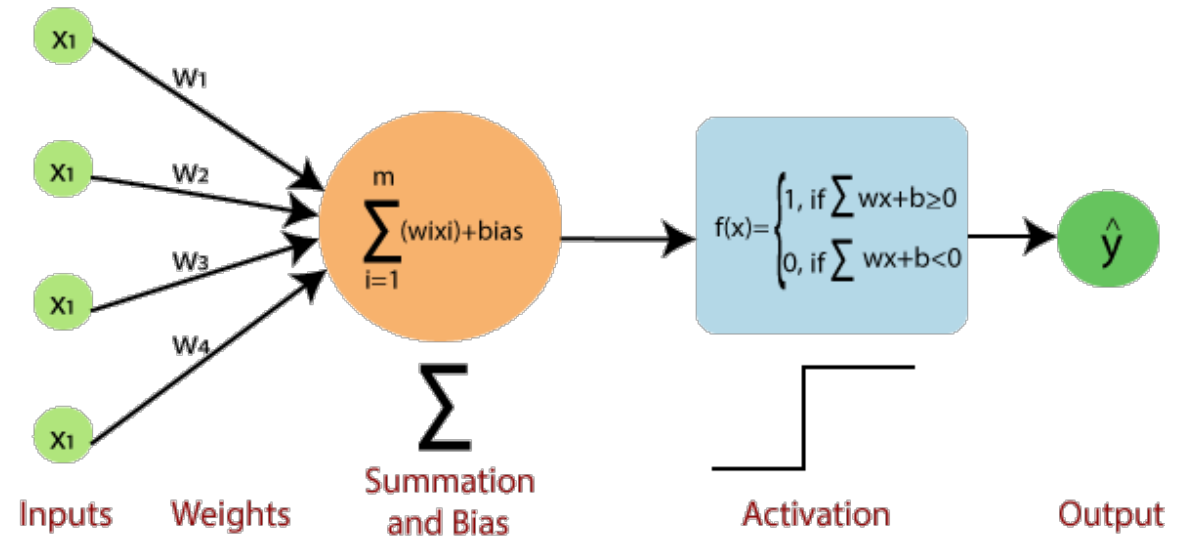
Rosenblatt's Perceptron (1958)

- **Learns from labels!**

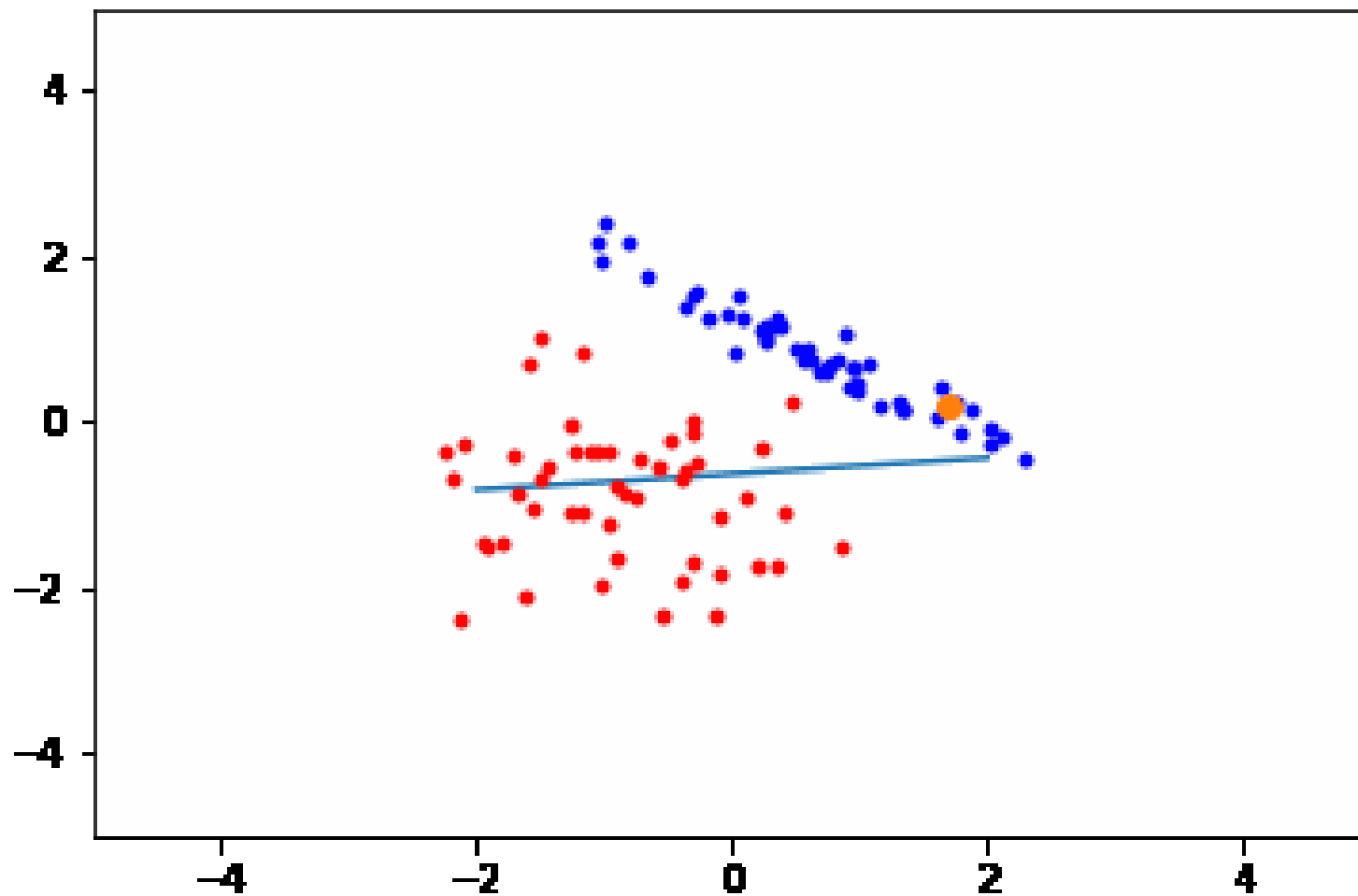


The Perceptron

- Input: feature vector $x \in \mathbb{R}^m$
- Label: $y \in \{0, 1\}$
- Model:
$$\hat{y} = \sigma(w^T x + b)$$
- **Update rule:** $\Delta w = \eta(y - \hat{y})x$
- **Intuition:**
 - w is an hyperplane
 - We include or exclude the input based on difference from the label



Perceptron learning: the line moves toward misclassified points

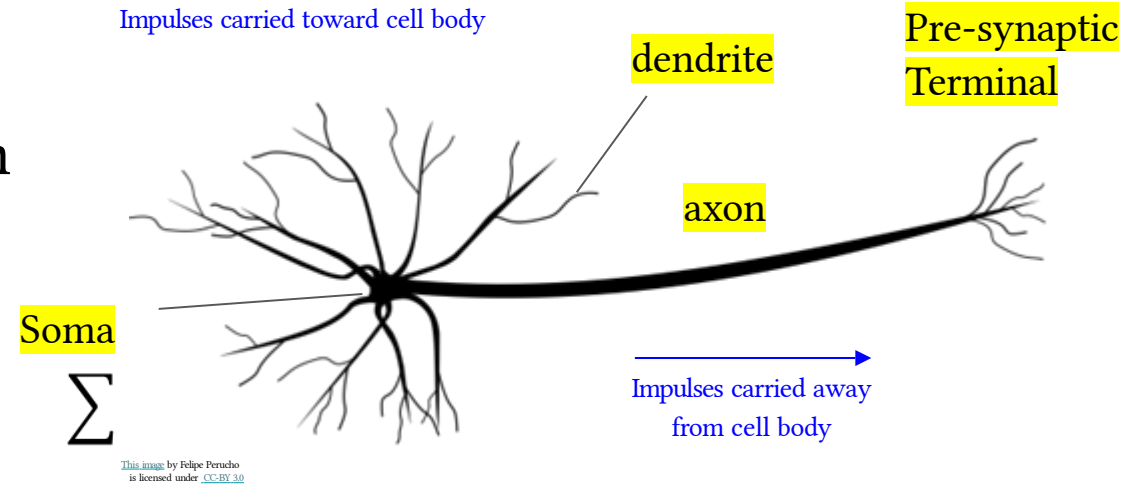


Human-like capabilities



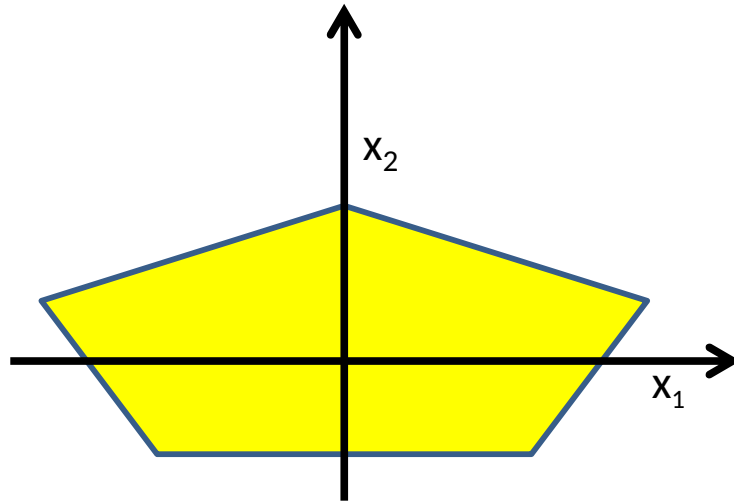
Real Neurons vs Perceptron

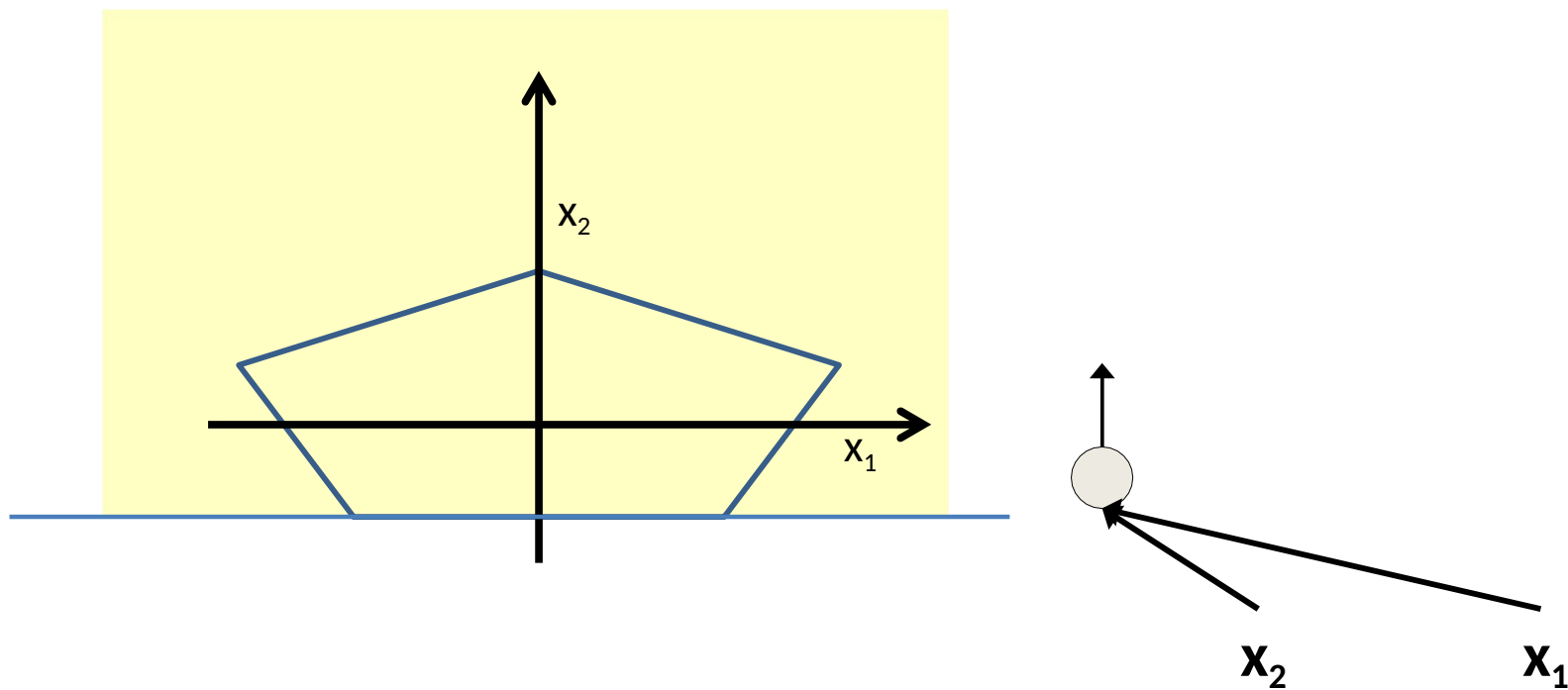
- Many neuron types
 - sensory, inhibitory, excitatory; each with unique roles.
- Dendrites compute non-linearly
 - amplify some signals, cancel others.
- Synapses are dynamic
 - weights change with timing, frequency, and chemicals.

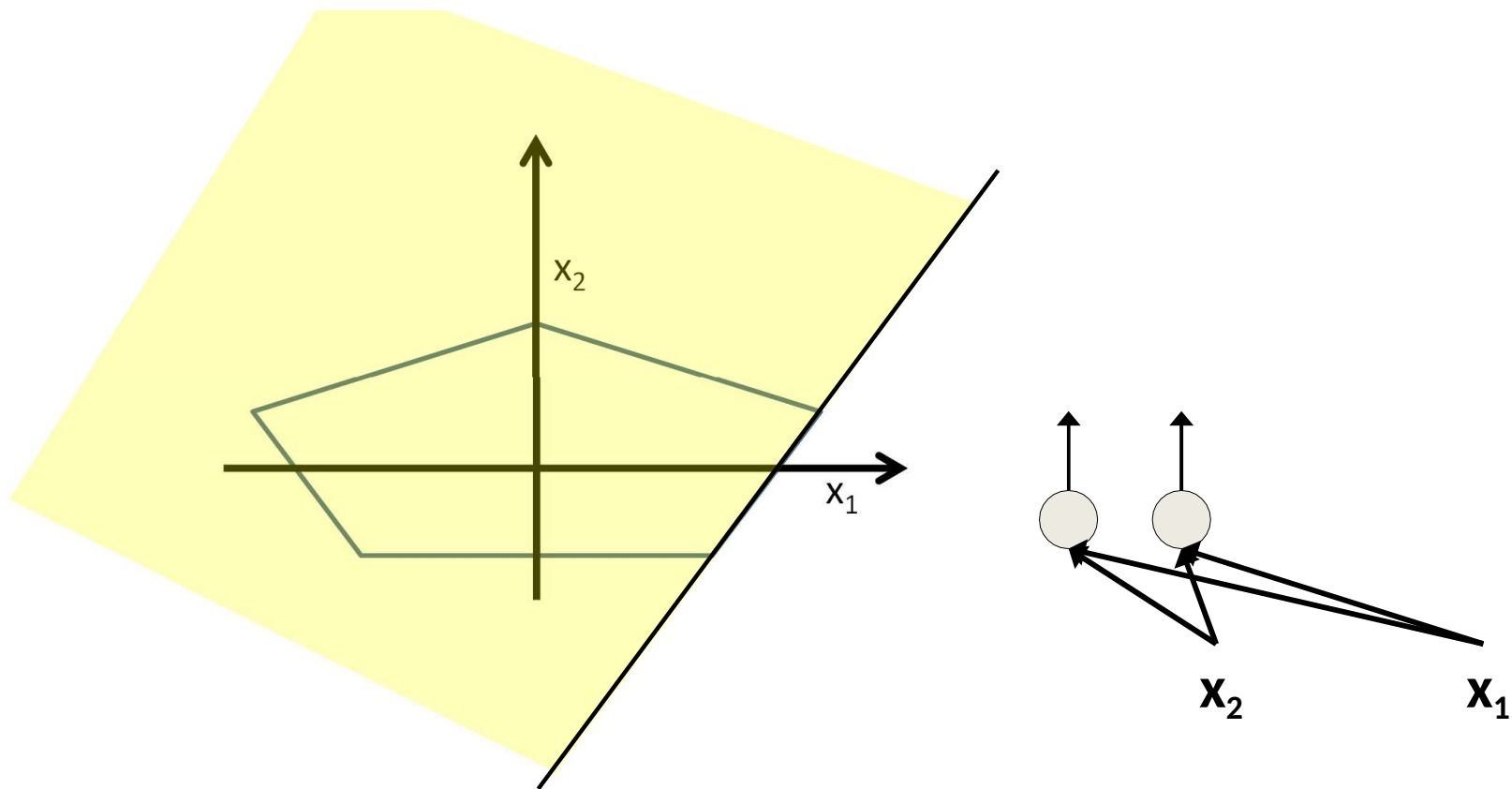


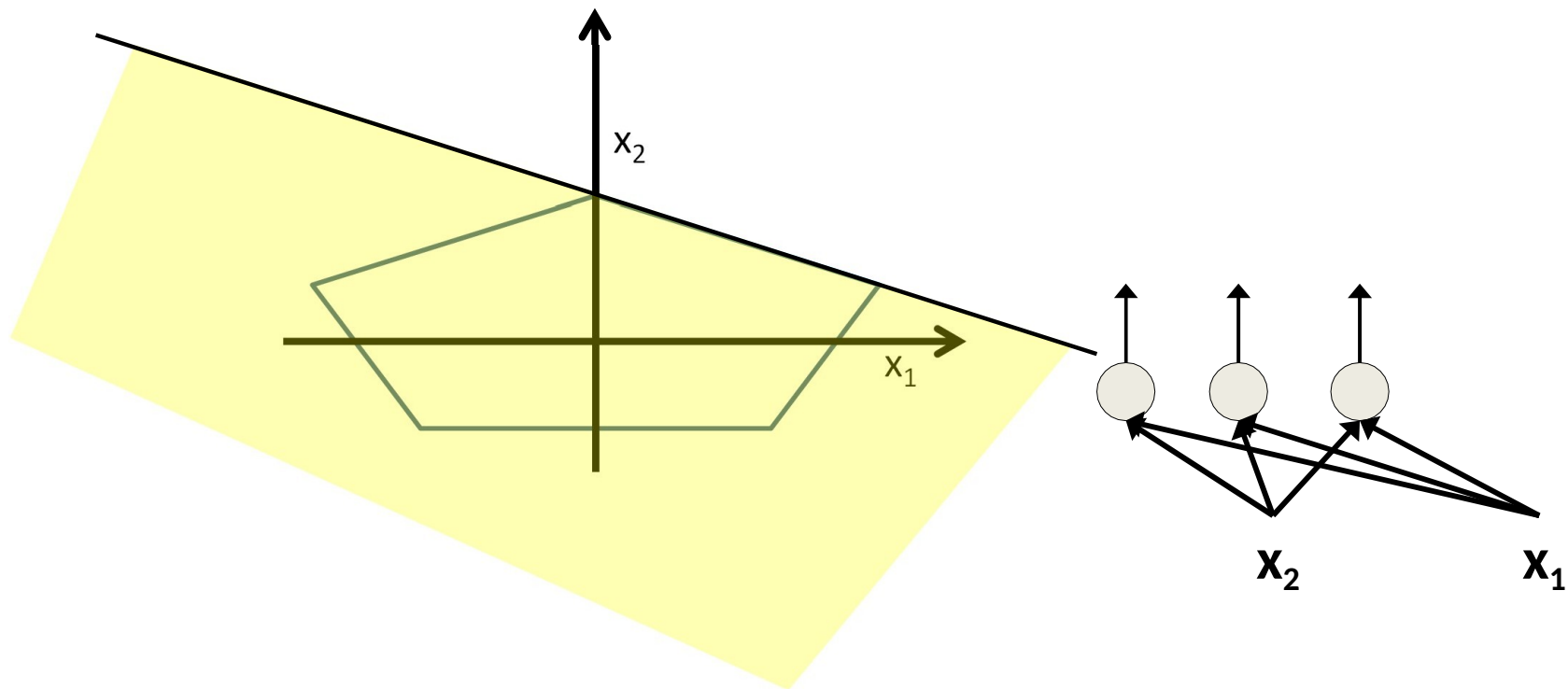
Composing “decision” boundaries

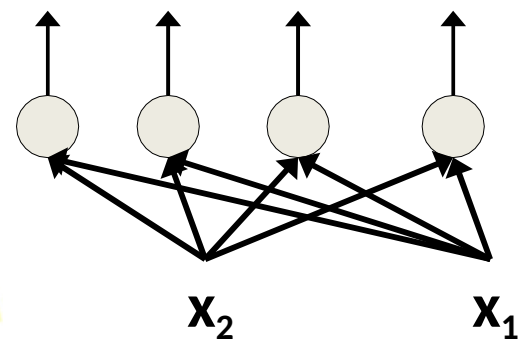
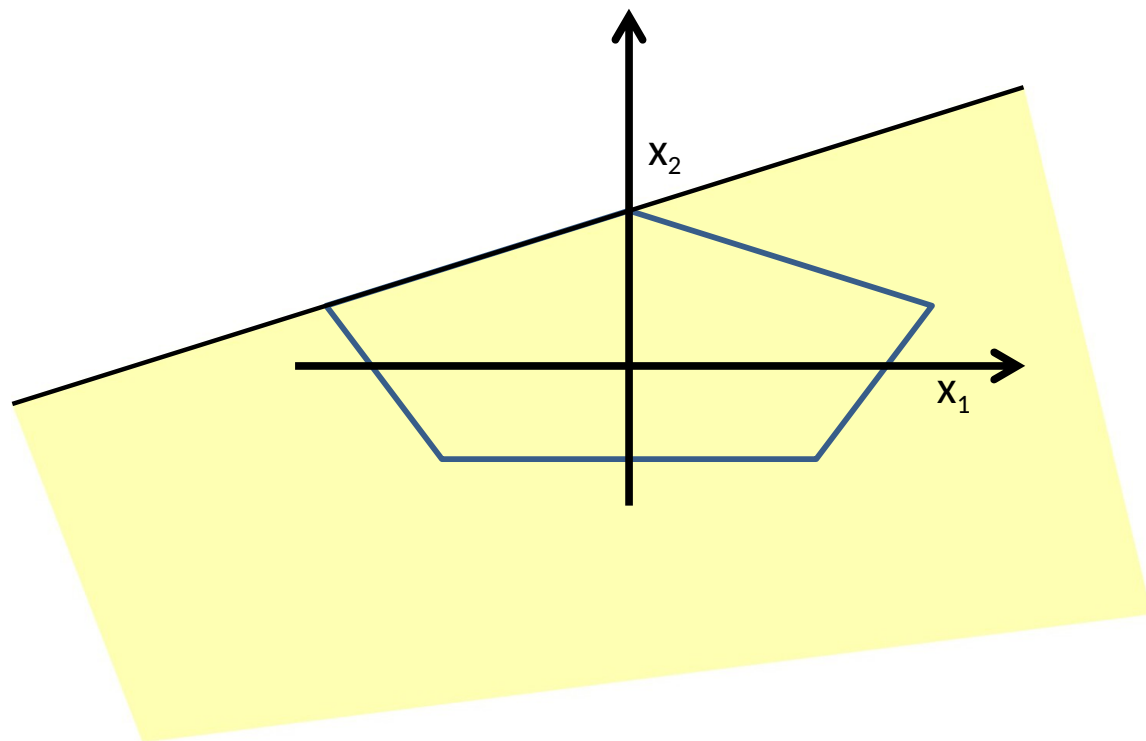
- Build a network of units with a single output that fires if the input is in the colored area

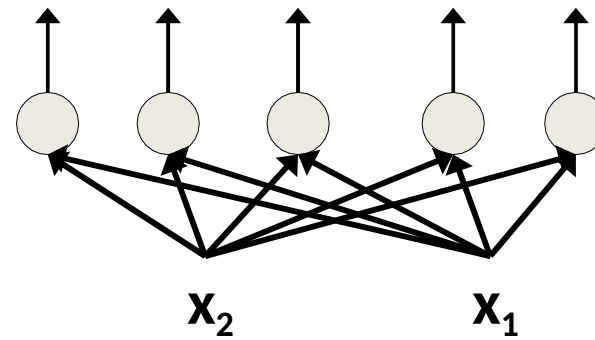
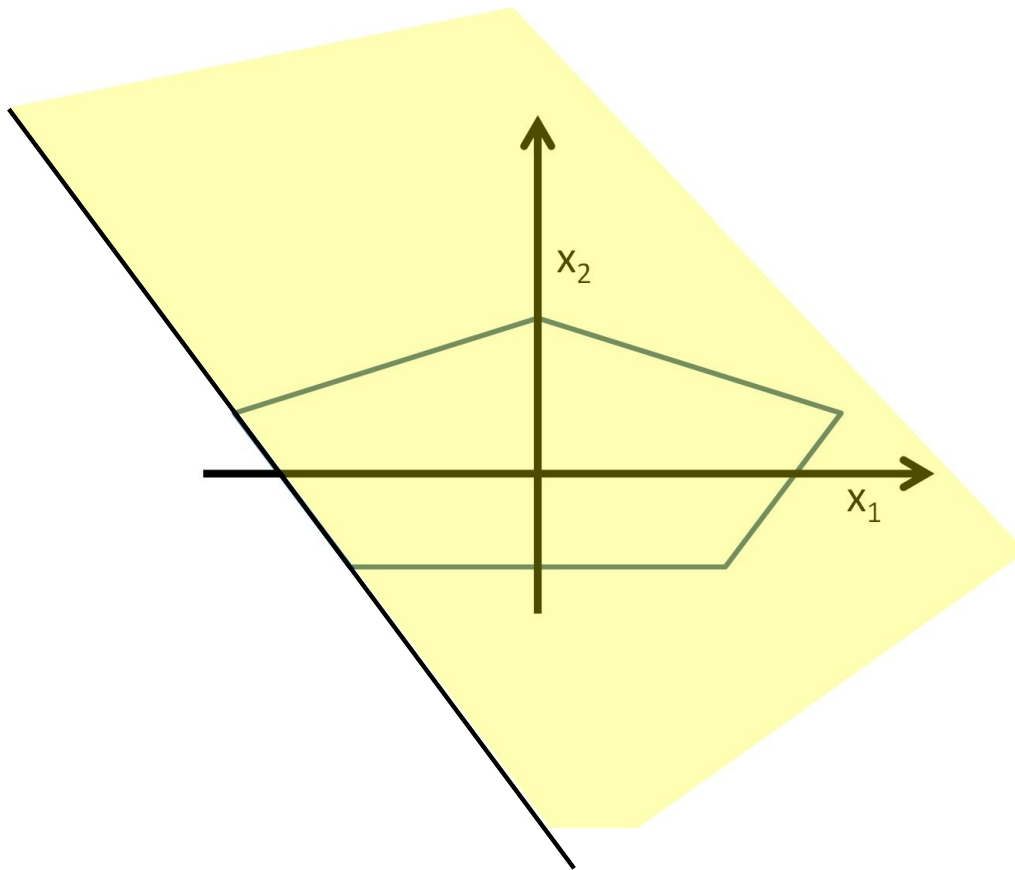


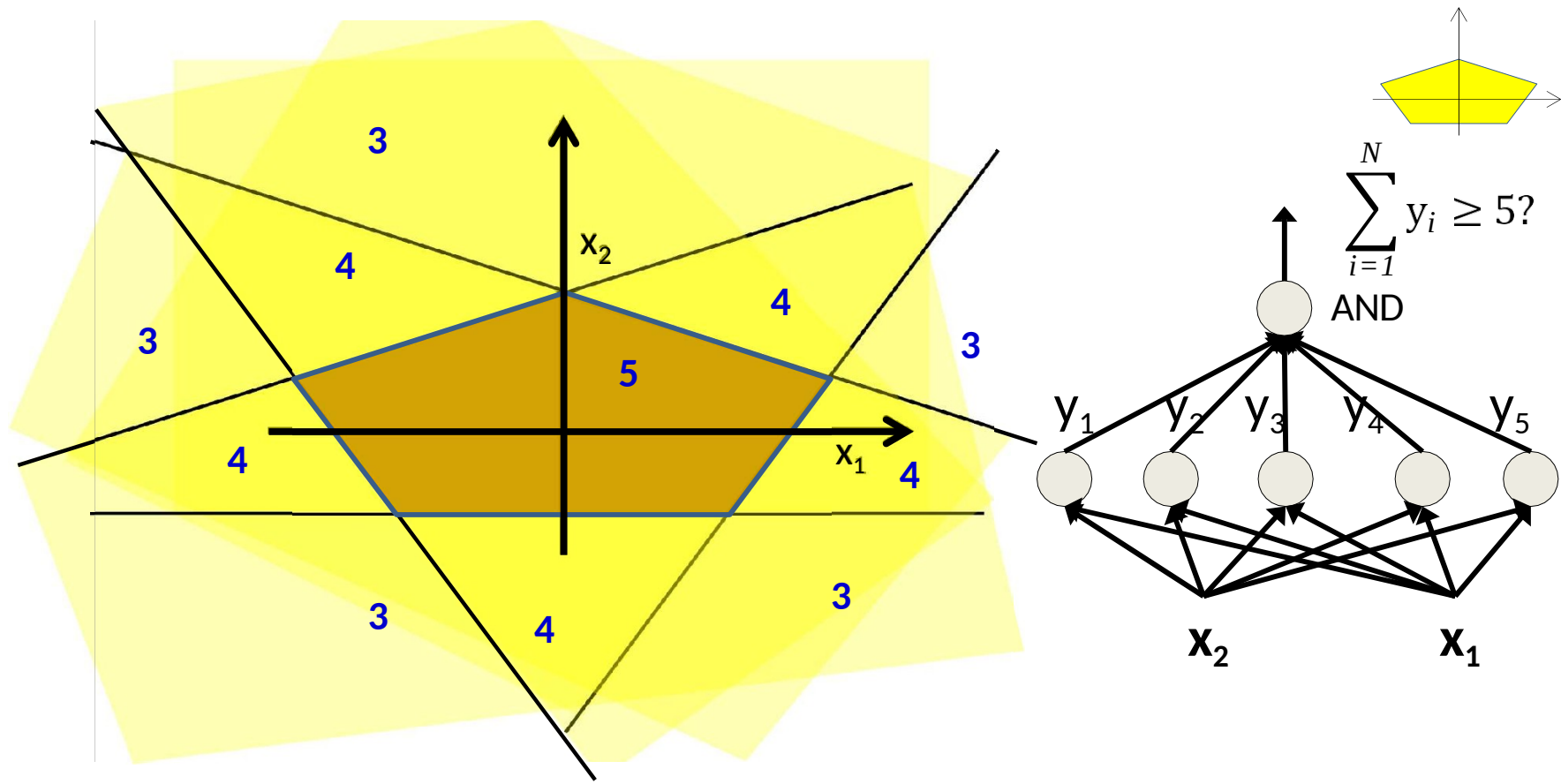






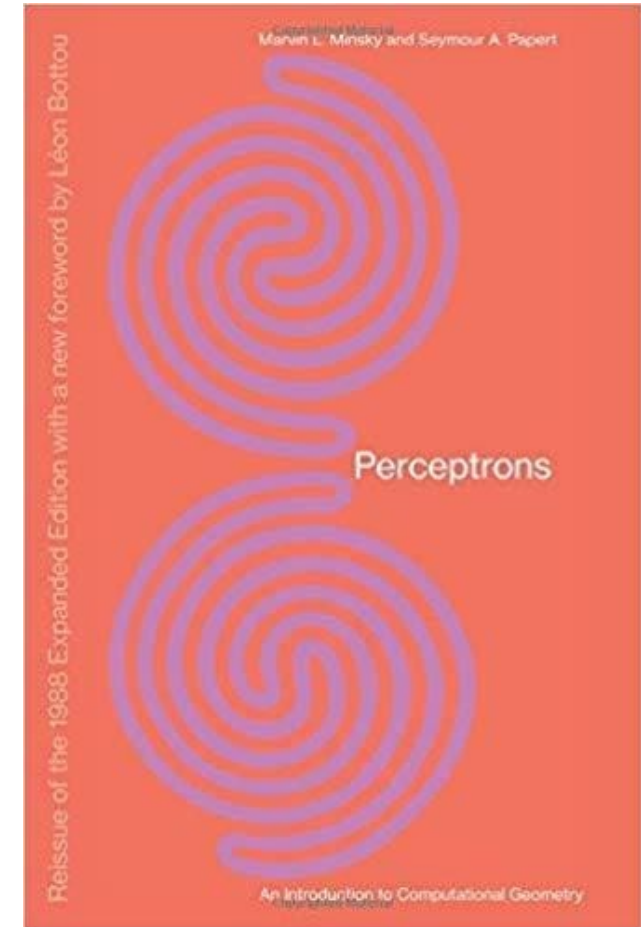
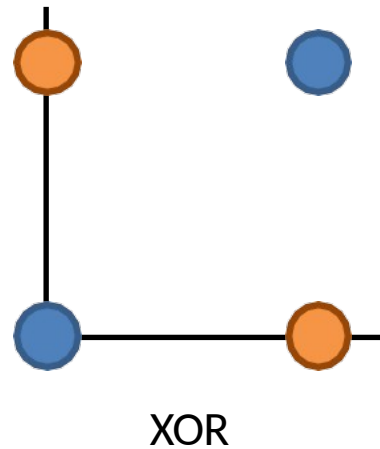






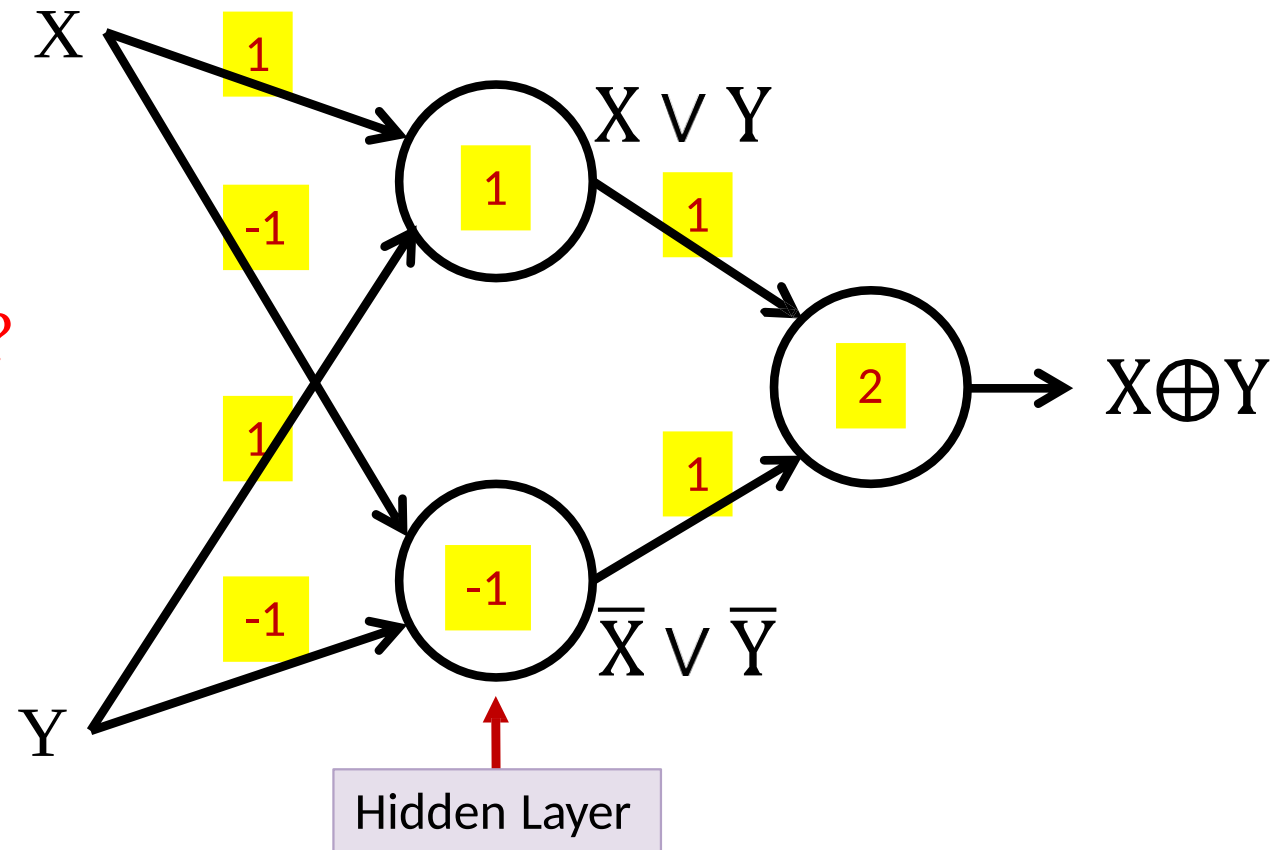
Minsky and Papert, 1969 (AI Winter)

- Showed that Perceptron could not learn the XOR function

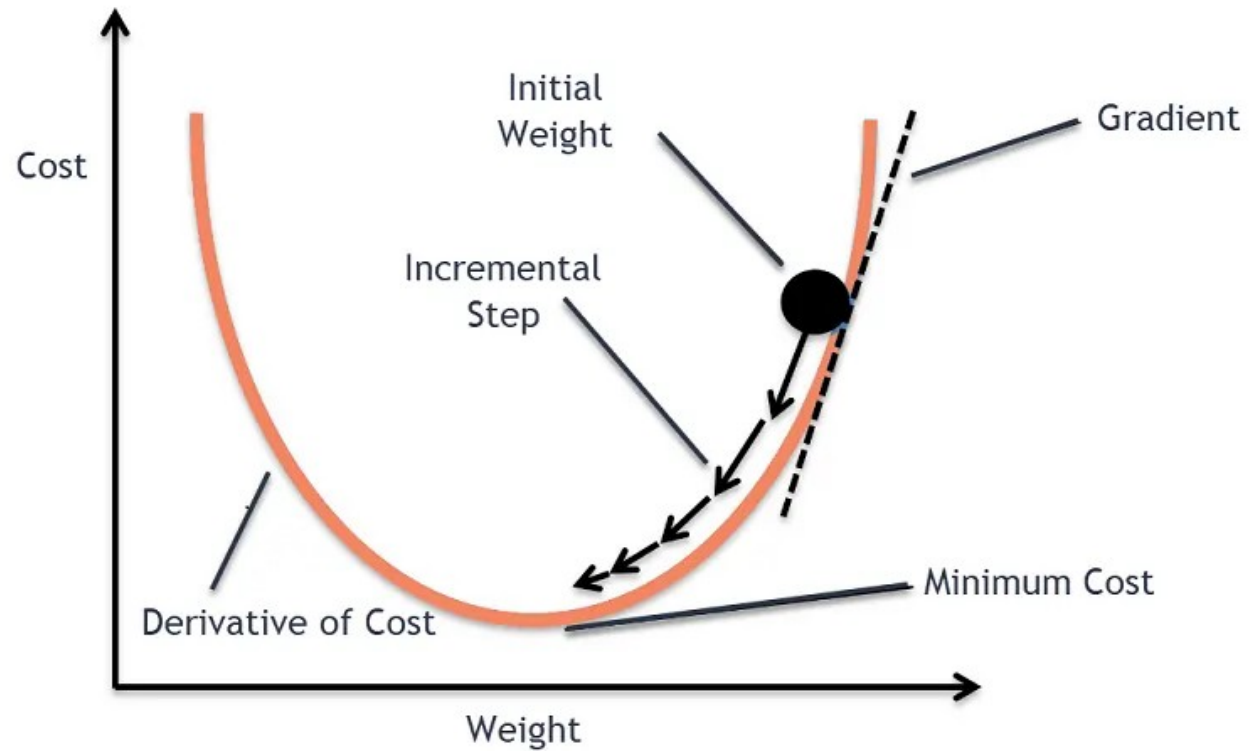
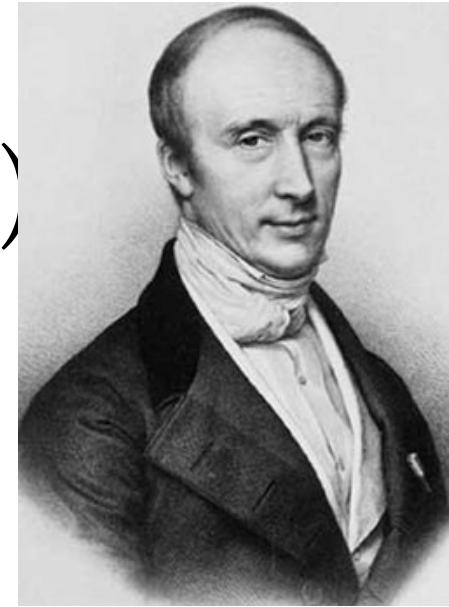


XOR and Multi-layer Perceptron (MLP)

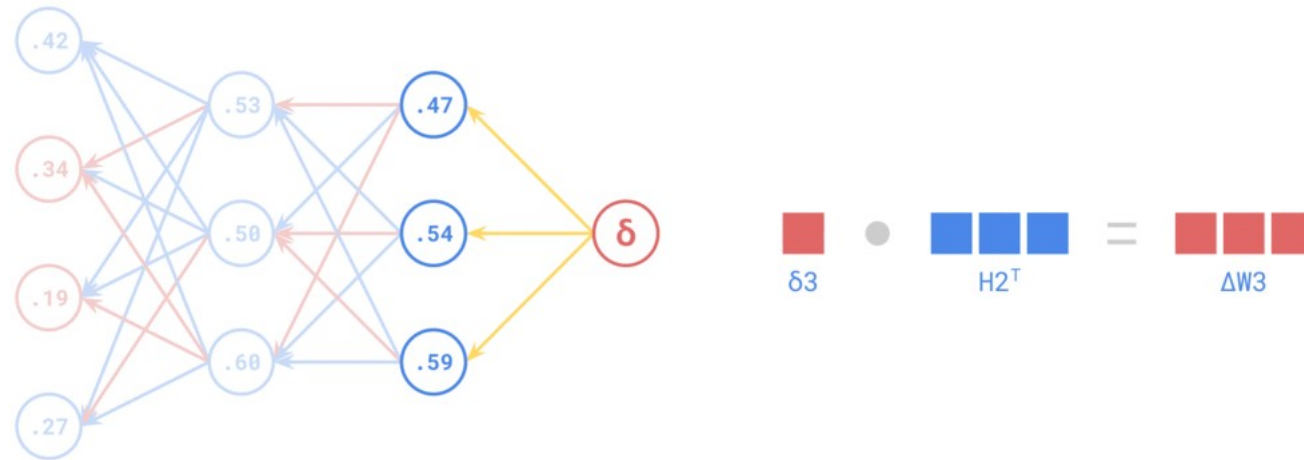
But how to train?



Cauchy and the Gradient Method (1847)



Backpropagation for MLP



Development of Backpropagation

- Optimization methods based on chain rule
 - Gradient Theory of Optimal Flight Paths [**Kelley 1960**]
- Generic and efficient derivations
 - Taylor expansion of the accumulated rounding error [**Linnainmaa 1976**]
- First connection to neural networks
 - **P. J. Werbos** Thesis **1974**
 - System Modeling and Optimization **1981**
- Popularize representation learning
 - Learning representations by back-propagating errors [DE. Rumelhart, **GE. Hinton**, RJ. Williams **1986**]



Paul Werbos

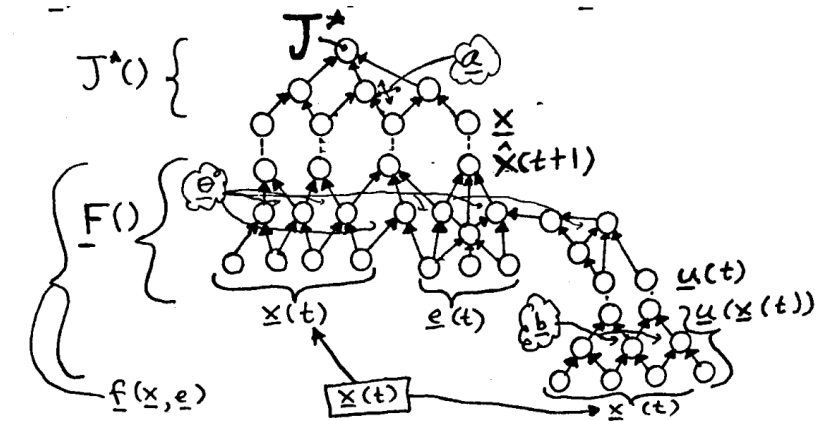
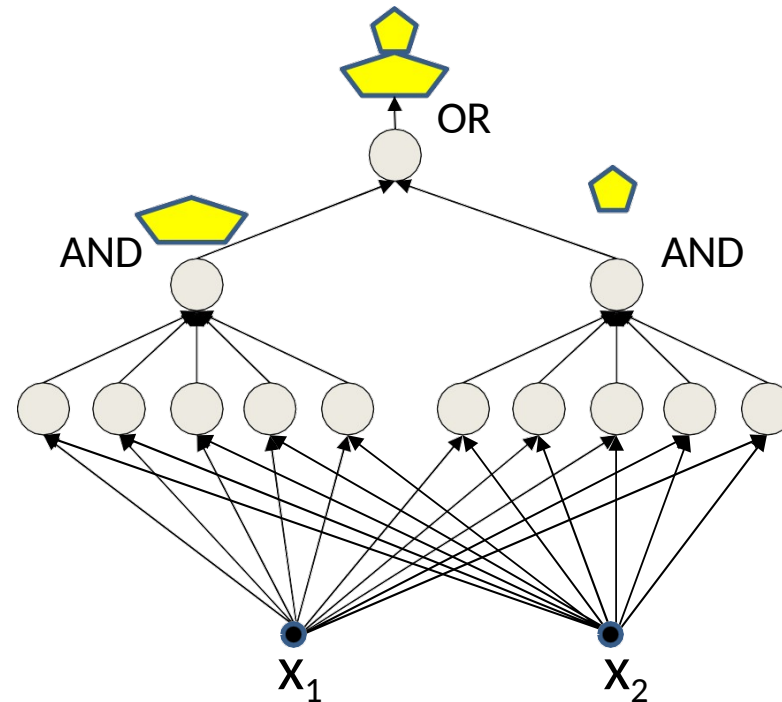
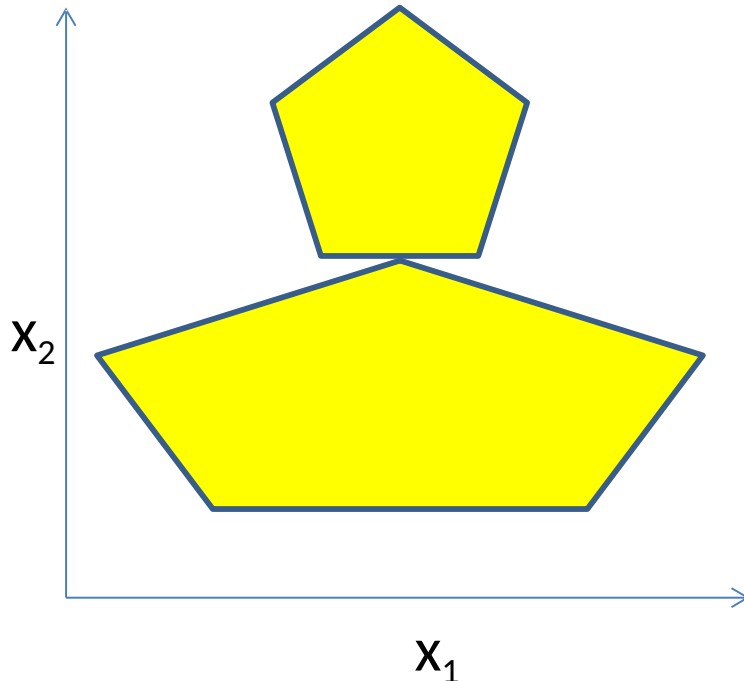


Figure 5: Realization of GDHP As a Triple Network to Make Decisions

Figure from System Modeling and Optimization

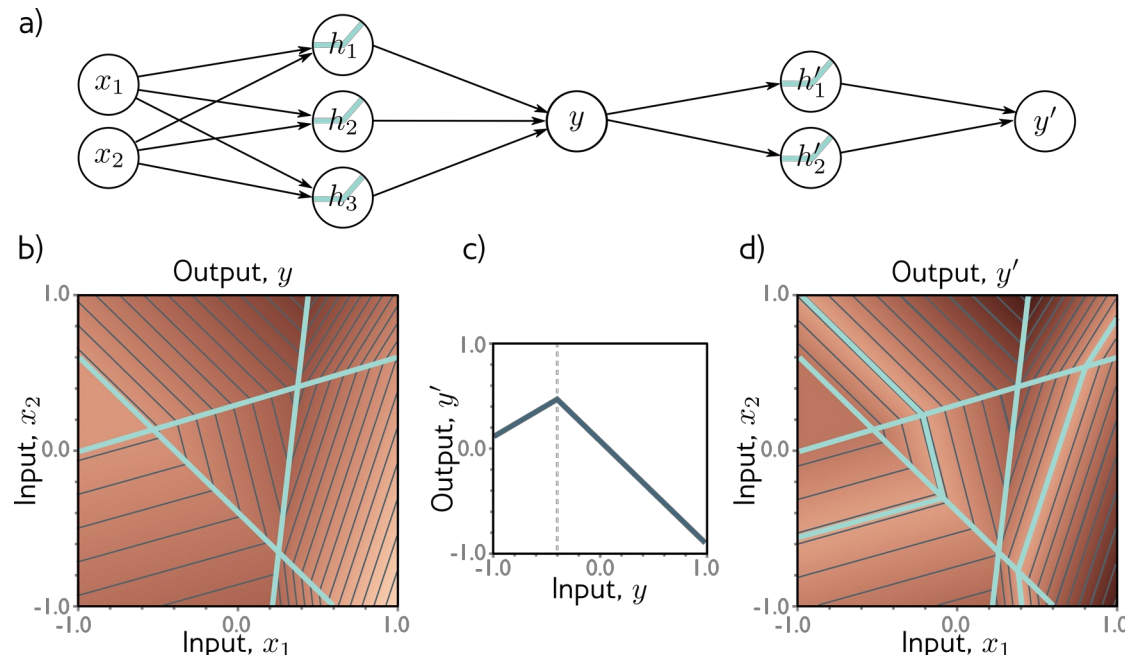
More complex decision boundaries

- Network to fire if the input is in the yellow area
 - “OR” two polygons
 - A third layer is required

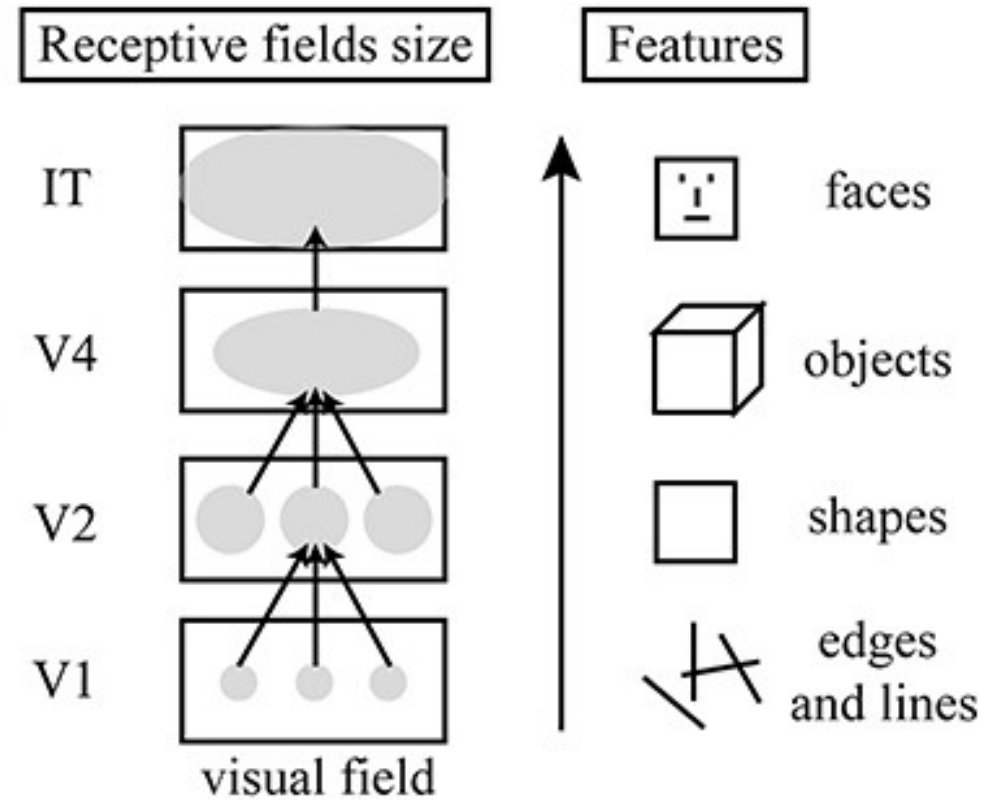
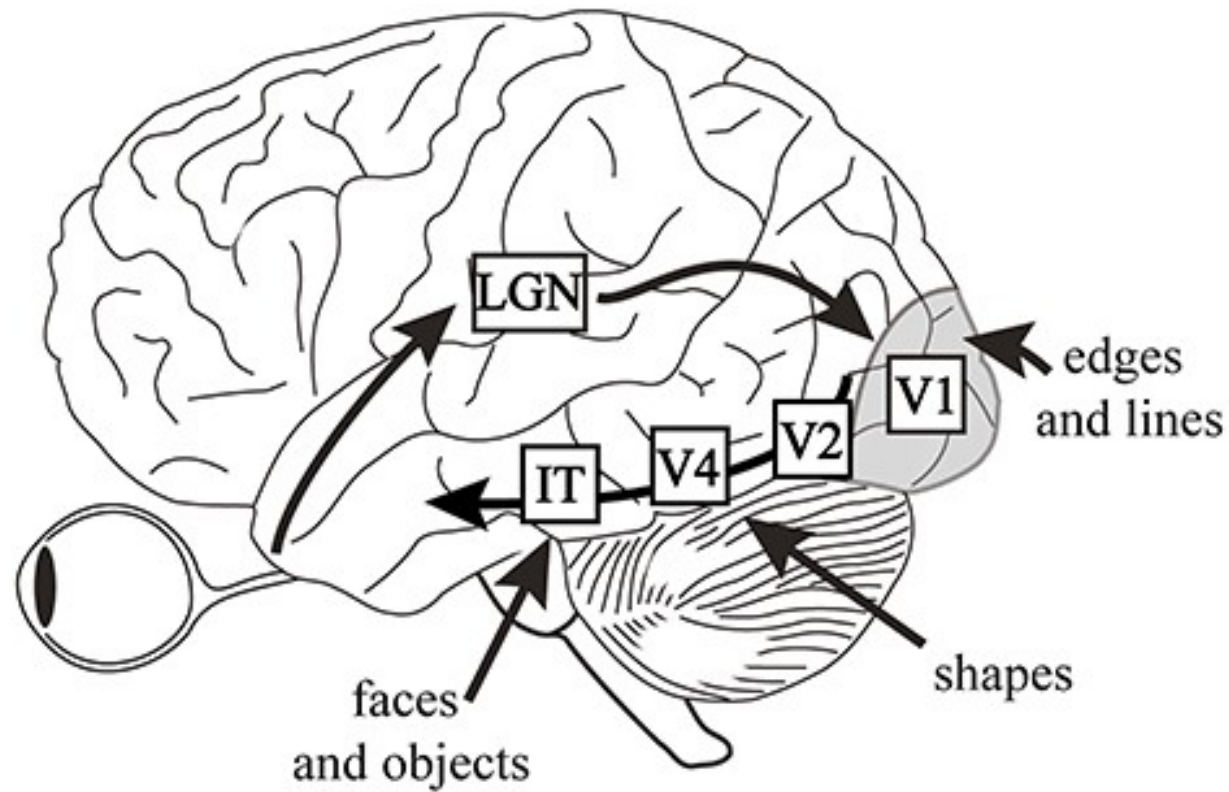


Depth efficiency

- Deep networks create many more regions per parameters
- But there are dependencies between them



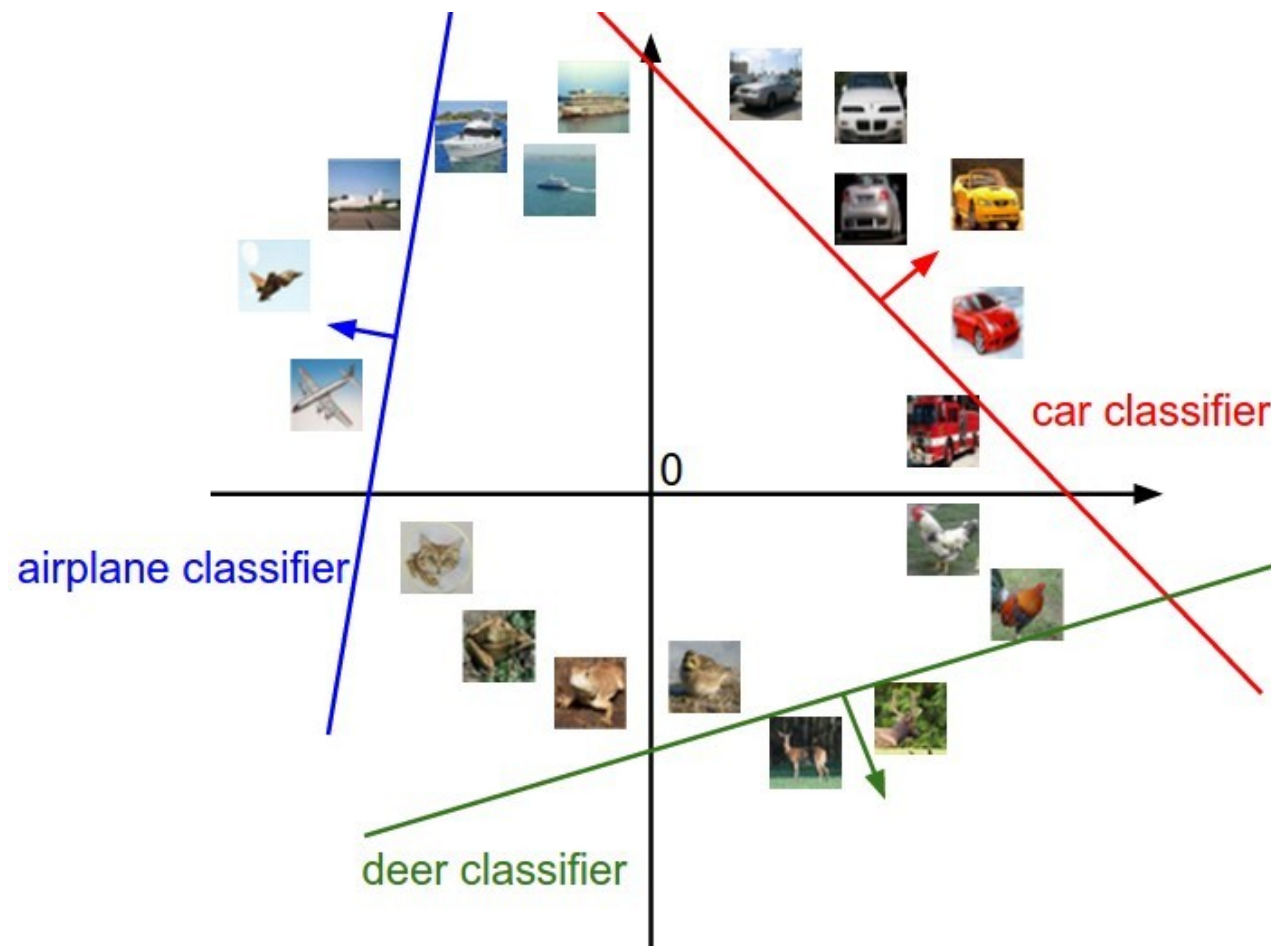
Deep learning and connection to the Visual Cortex



Hubel & Wiesel Experiment (1959)

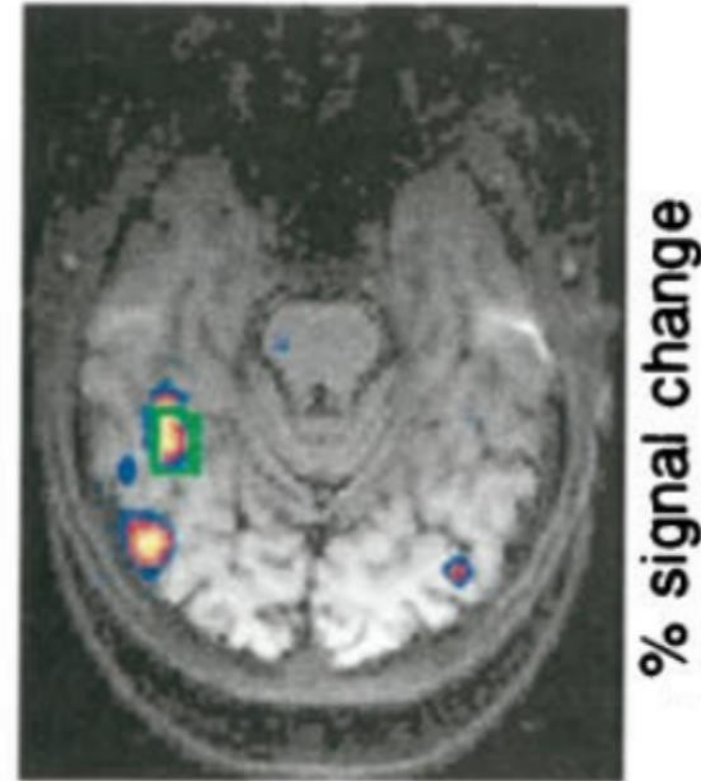


Image Classification?



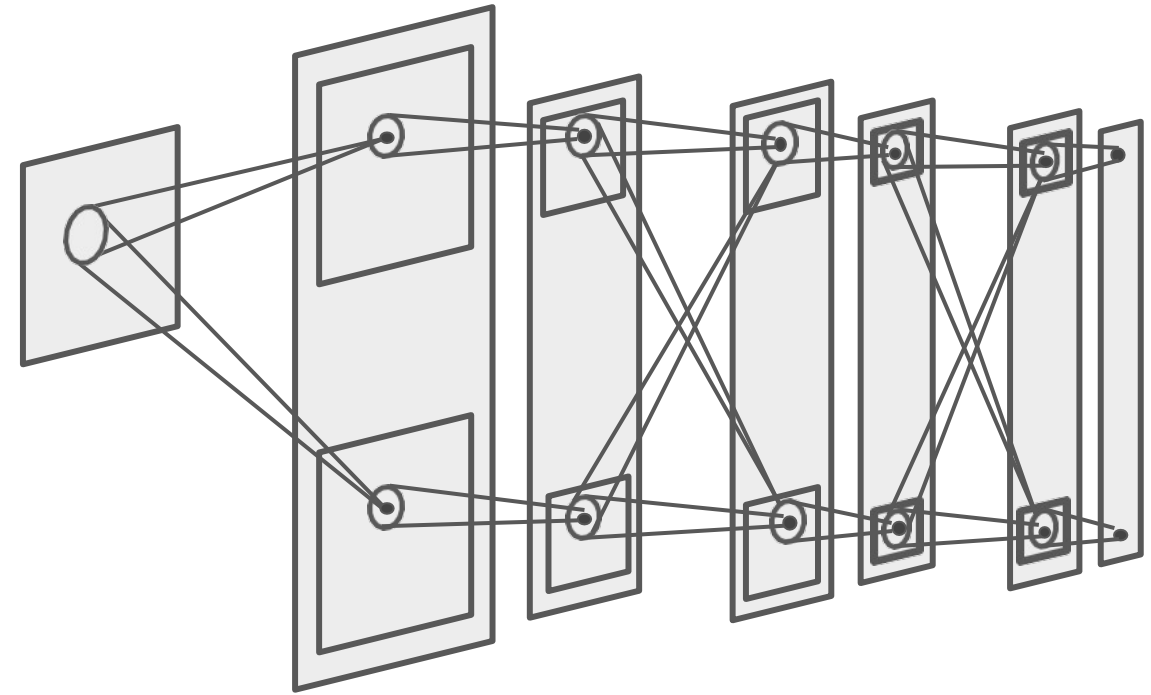
Neural correlation to abstract concepts (Kanwisher et al., 1997)

Faces > Houses



Neocognitron: Fukushima, 1980

- Computational model the visual system, directly inspired by Hubel and Wiesel's hierarchy of complex and simple cells
- Interleaved simple cells (convolution) and complex cells (pooling)
- **No practical training algorithm**



Running example

- For example, let's take the filter

1	0	1
0	1	0
1	0	1

1x1	1x0	1x1	0	0
0x0	1x1	1x0	1	0
0x1	0x0	1x1	1	1
0	0	1	1	0
0	1	1	0	0

4		

Known Filters



0	0	0	0	0
0	0	-1	0	0
0	-1	5	-1	0
0	0	-1	0	0
0	0	0	0	0

Sharpen



0	0	0	0	0
0	1	1	1	0
0	1	1	1	0
0	1	1	1	0
0	0	0	0	0

Blur



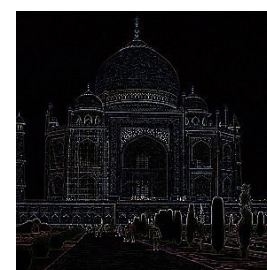
	0	0	0	
	-1	1	0	
	0	0	0	

Edge enhance



	0	1	0	
	1	-4	1	
	0	1	0	

Edge detect



	-2	-1	0	
	-1	1	1	
	0	1	2	

Emboss

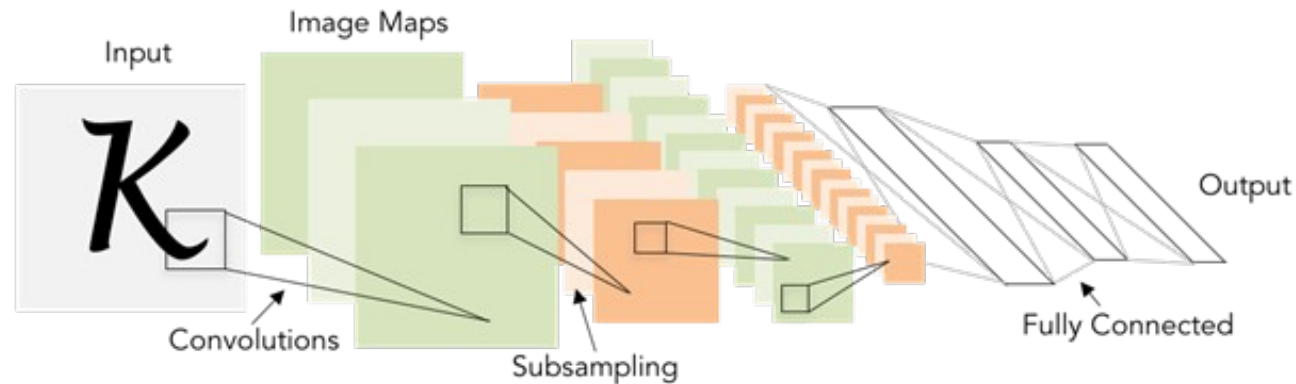


LeNet-1 reads zip codes (LeCun et al., 1989)



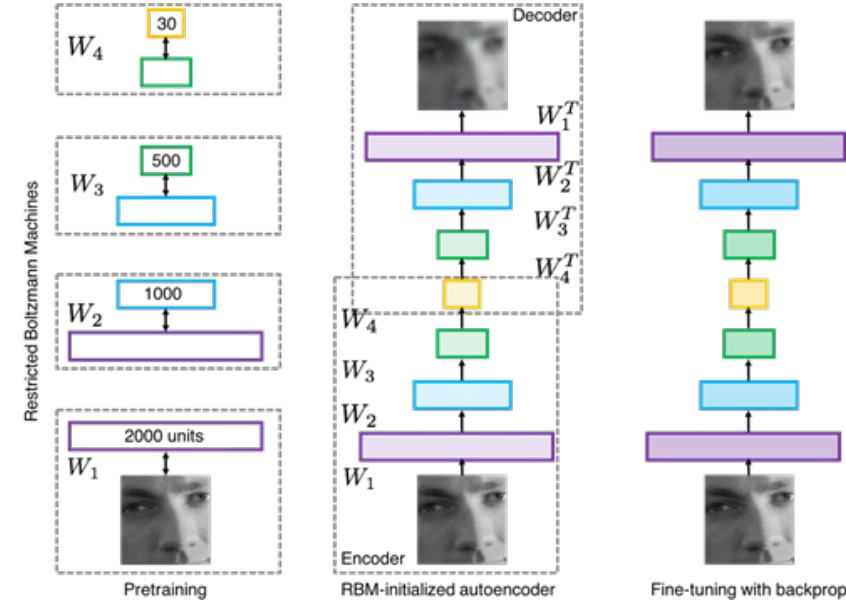
LeNet-5 (LeCun et al., 1998)

- Applied backprop algorithm to a Neocognitron-like architecture
- Learned to recognize handwritten digits
- Deployed commercially by NCR and AT&T to read bank checks
- Very similar to our modern convolutional networks!



2000s: “Deep Learning”

- Exists but not a mainstream research topic
 - No large-scale benchmark (ImageNet: 2009)
 - No large datasets, little compute

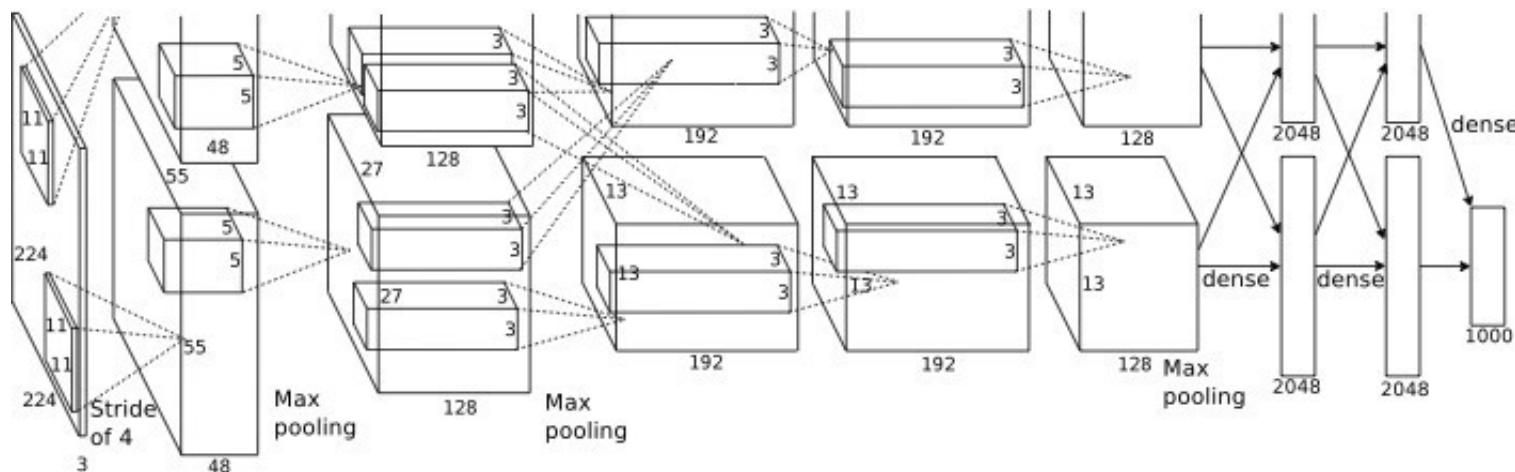


Hinton and Salakhutdinov, 2006

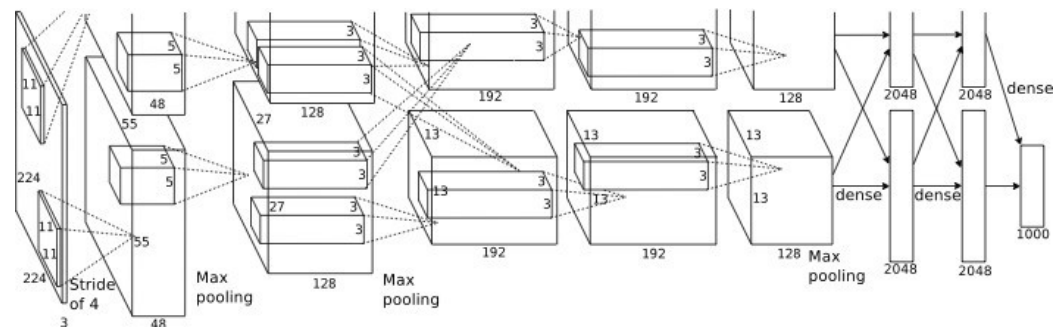
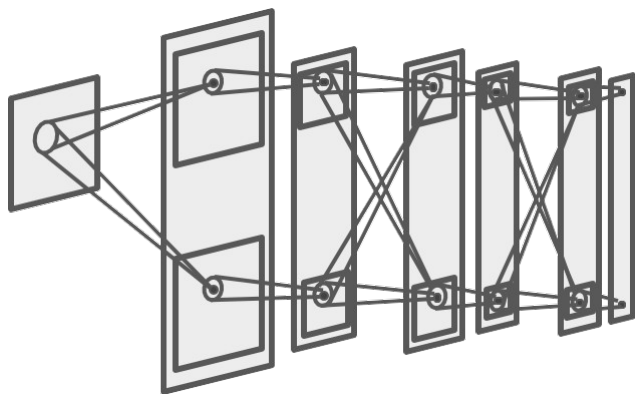
AlexNet (2012)



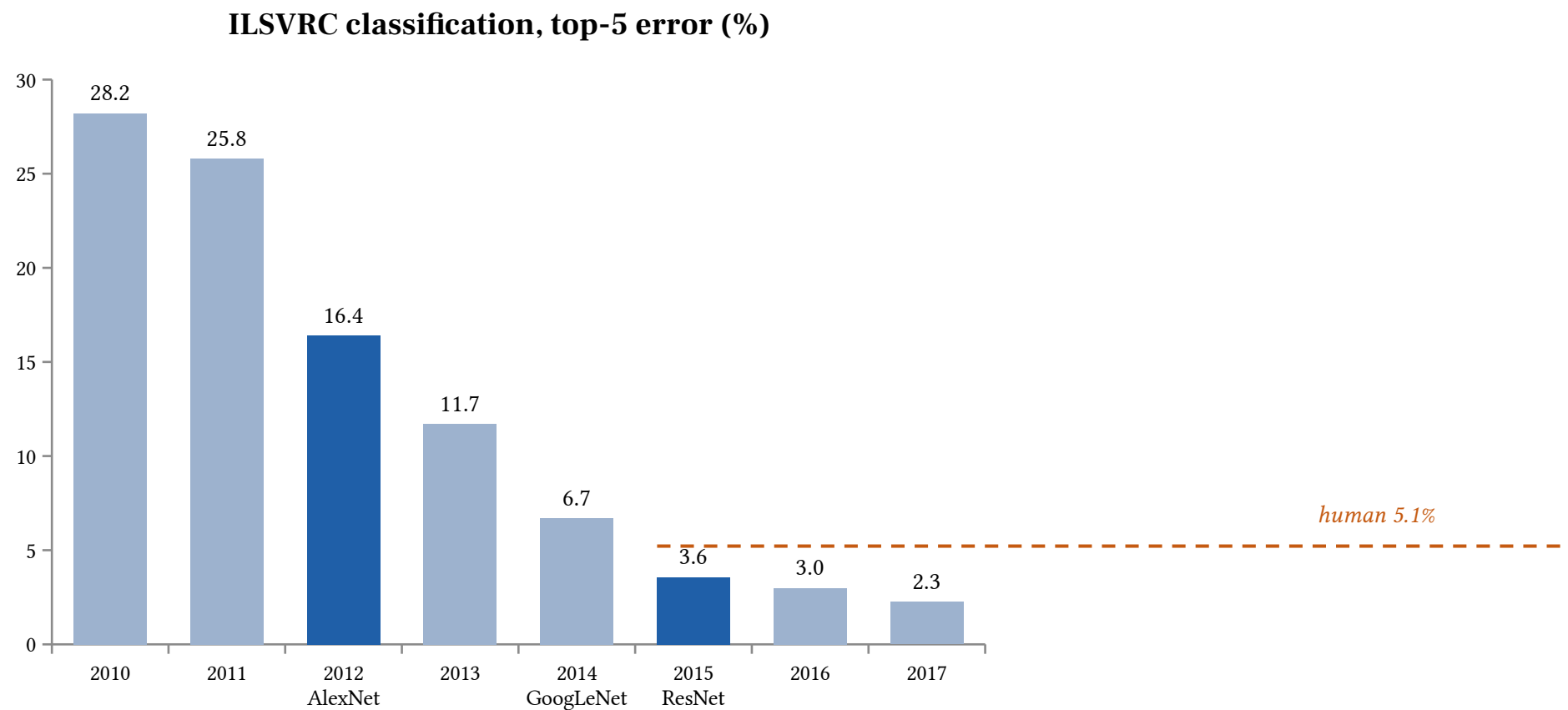
- Significant improvements on large-scale classification challenge
- Very efficient (ReLU activations, GPU-based optimization)
- Deep learning becomes mainstream



AlexNet vs. Neocognitron: 32 years apart



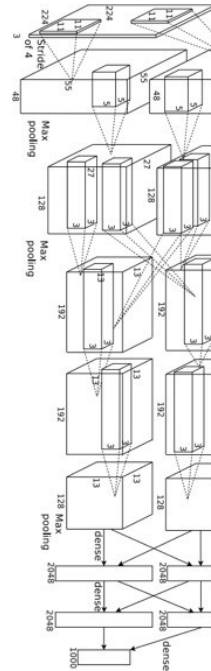
ImageNet (2009) and the ILSVRC challenge



Deeper every year: $8 \rightarrow 22 \rightarrow 152$ layers

Year 2012

SuperVision

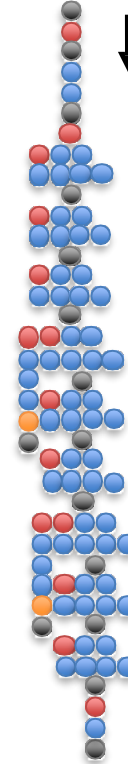


[Krizhevsky 2012]

Figure copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012. Reproduced with permission.

Year 2014

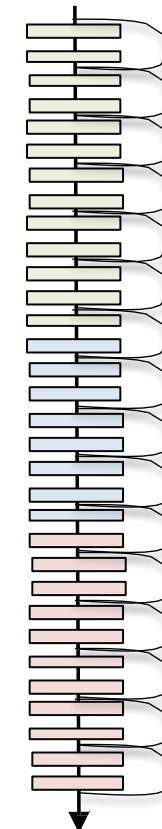
GoogLeNet



[Szegedy 2014]

Year 2015

ResNet



[He 2015]

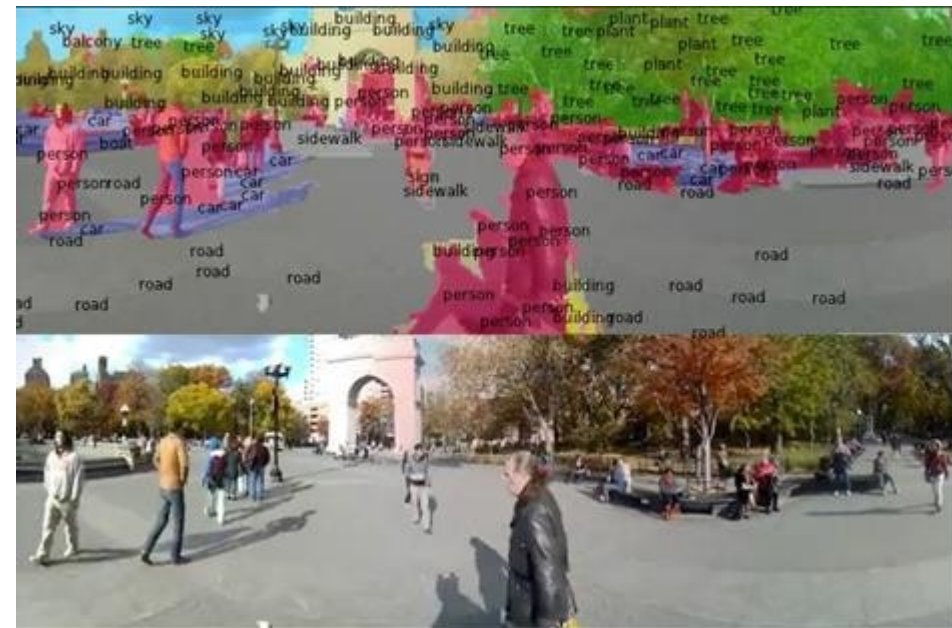
Object detection and segmentation (2012–2015)

Object Detection



Ren, He, Girshick, and Sun, 2015

Image Segmentation

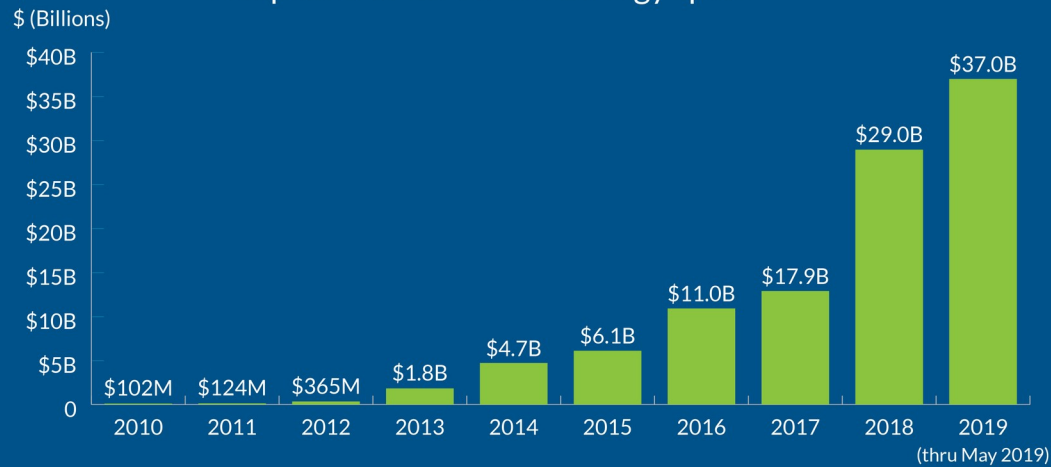


Farabet et al, 2012

Autonomous driving investments

Never before in the automotive industry have we seen this level of investment in new technologies...

Total cumulative funding (USD) for companies in the AV technology space



*Source: Crunchbase

...and AV investments are poised to set a new record in 2019!



Think like a robot, perceive like a human.

Turing Award for deep learning (2018)



[This image is CC0 public domain](#)



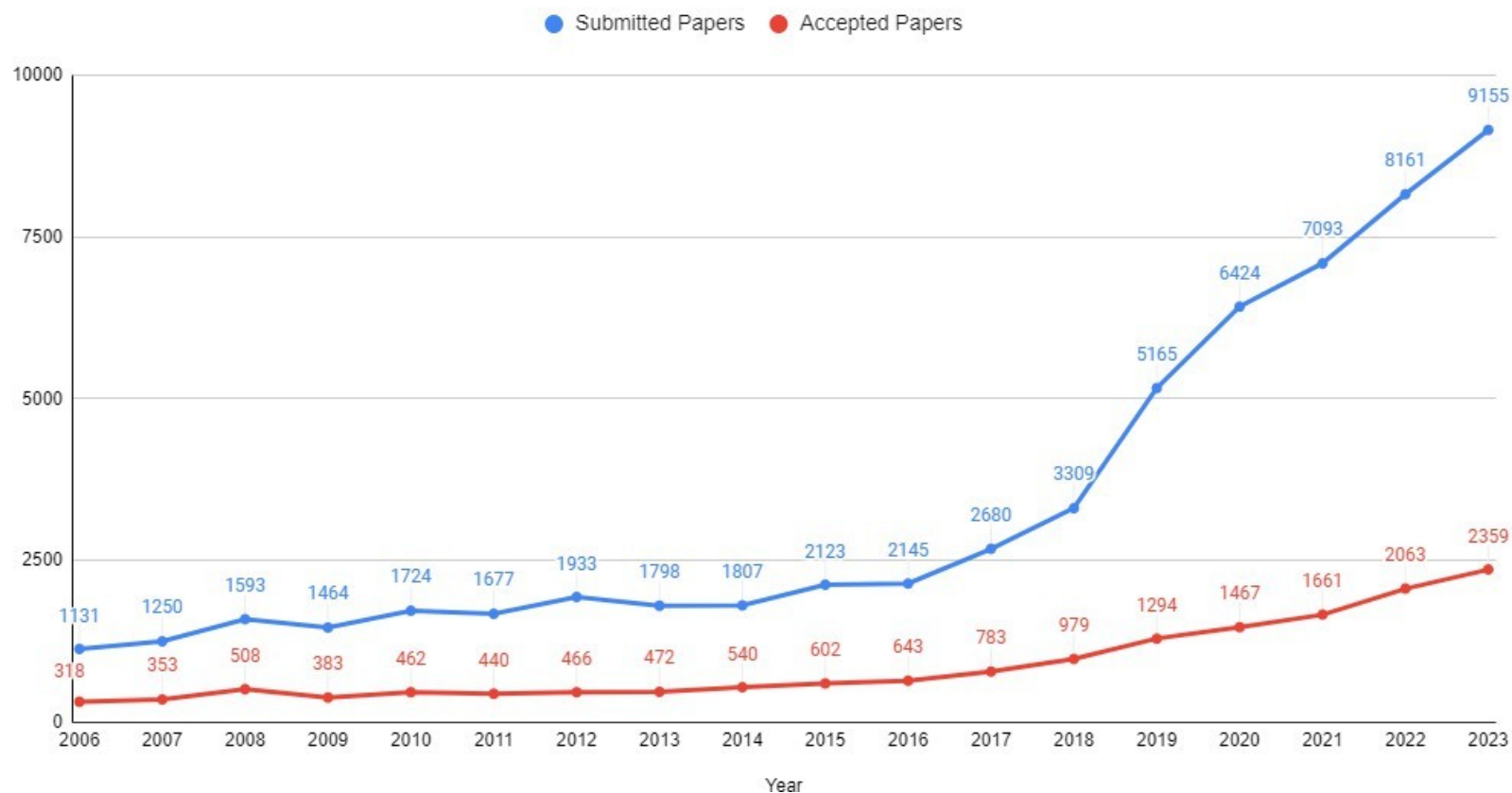
[This image is CC0 public domain](#)



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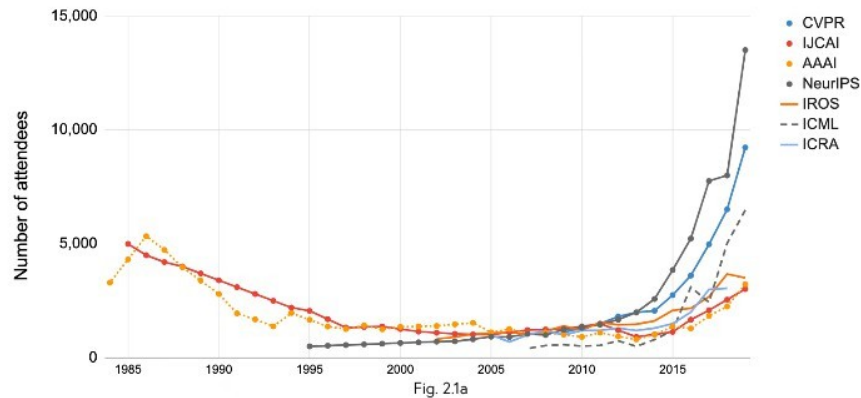
Deep learning explosion in academia

CVPR Conference Paper Statistics



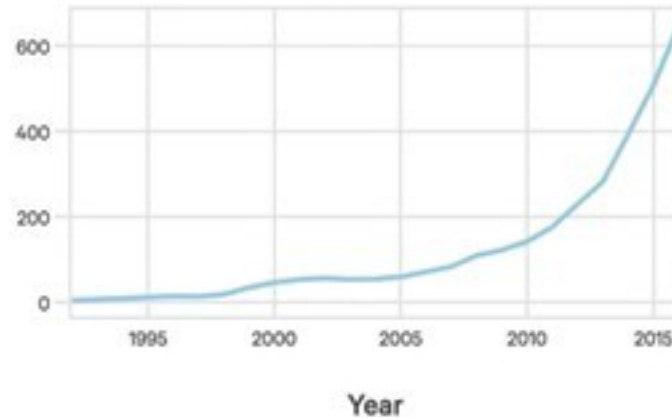
AI's explosive growth & impact

Attendance at large conferences (1984-2019)
Source: Conference provided data.



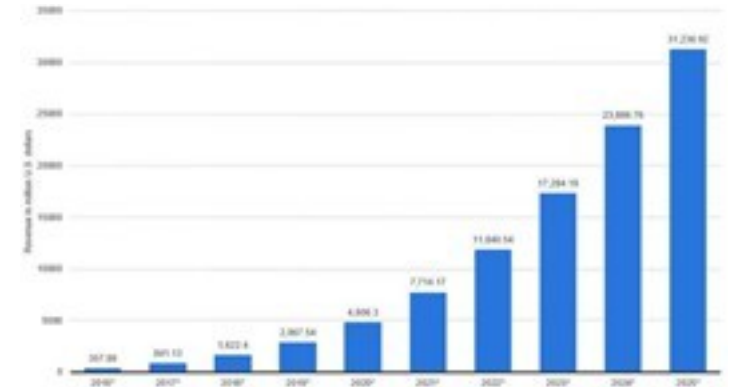
**Number of attendance
At AI conferences**

Source: The Gradient



**Startups Developing AI
Systems**

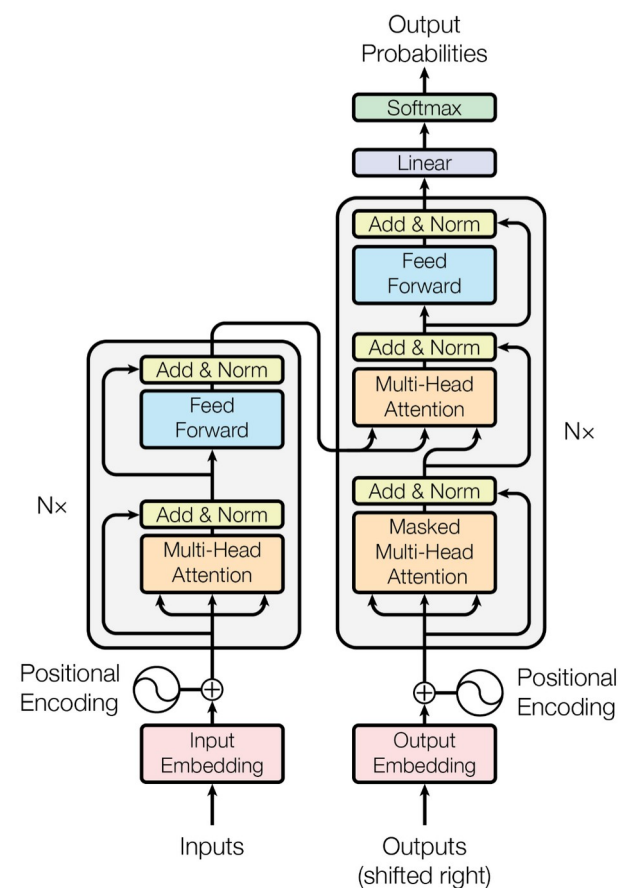
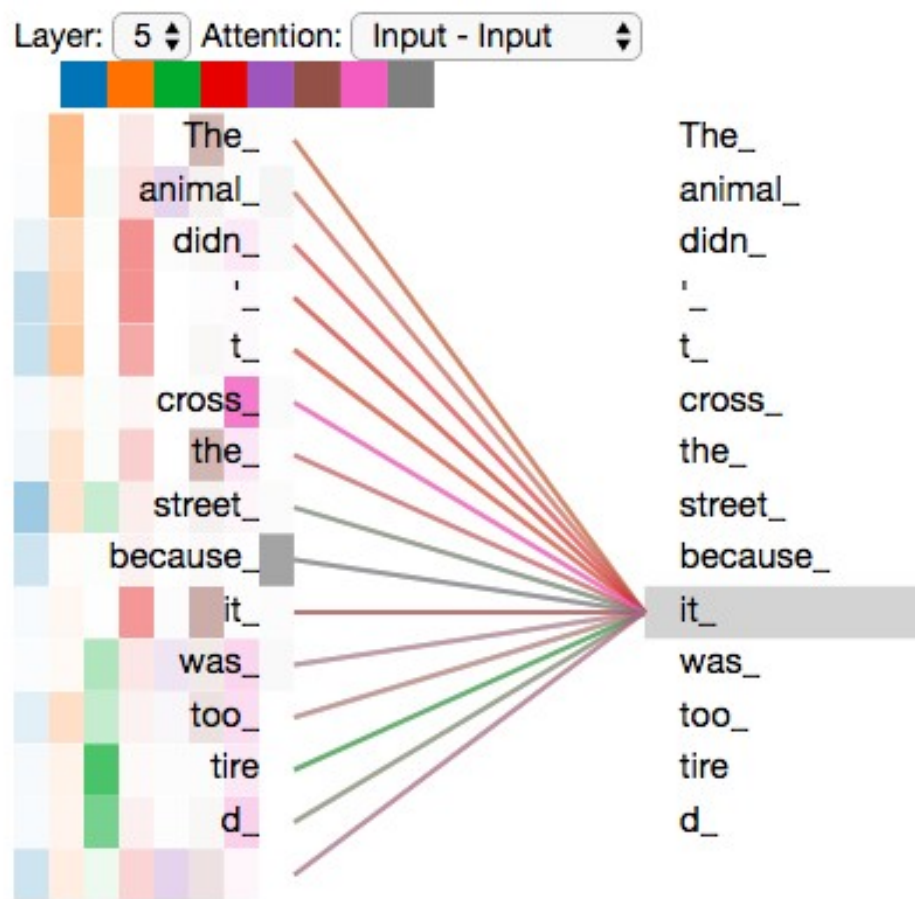
Source: Crunchbase, VentureSource, Sand
Hill Econometrics



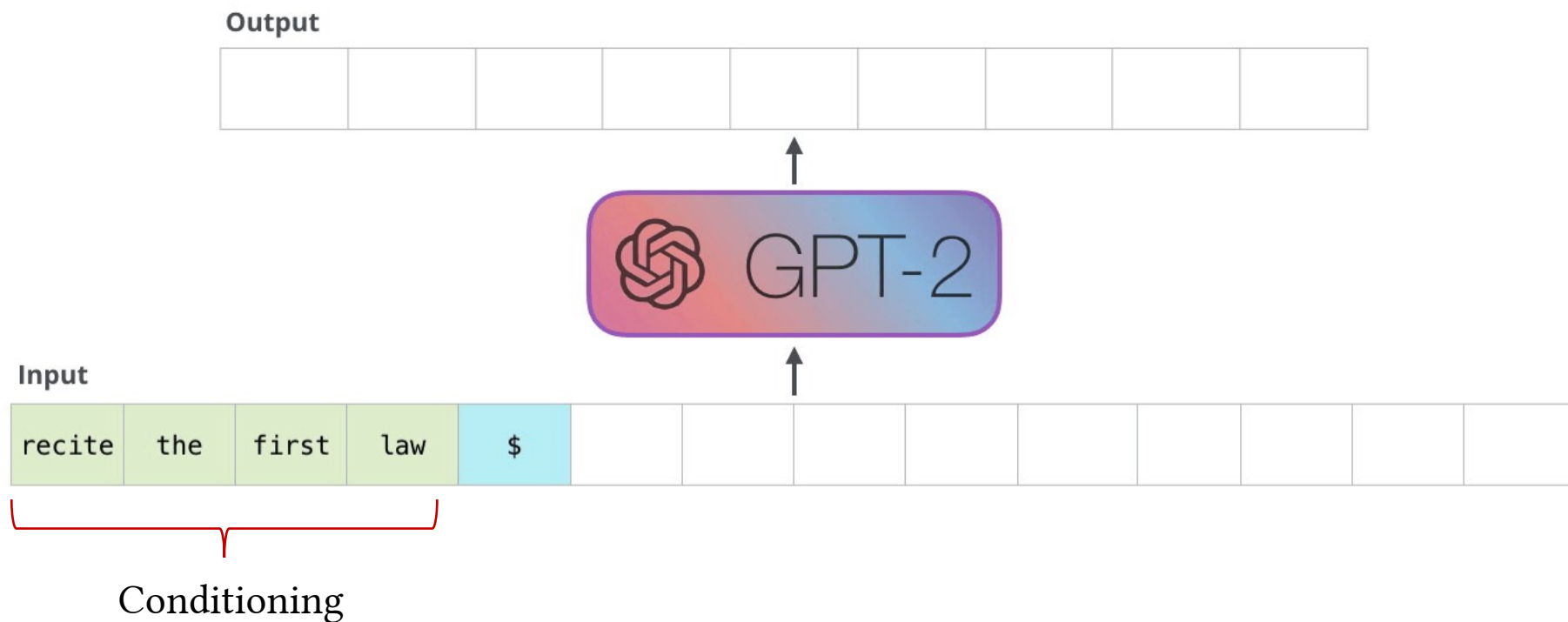
**Enterprise Application AI
Revenue**

Source: Statista

Attention is all you need (2017)



Large Language Models (2019)



The world has changed: ChatGPT (2022)

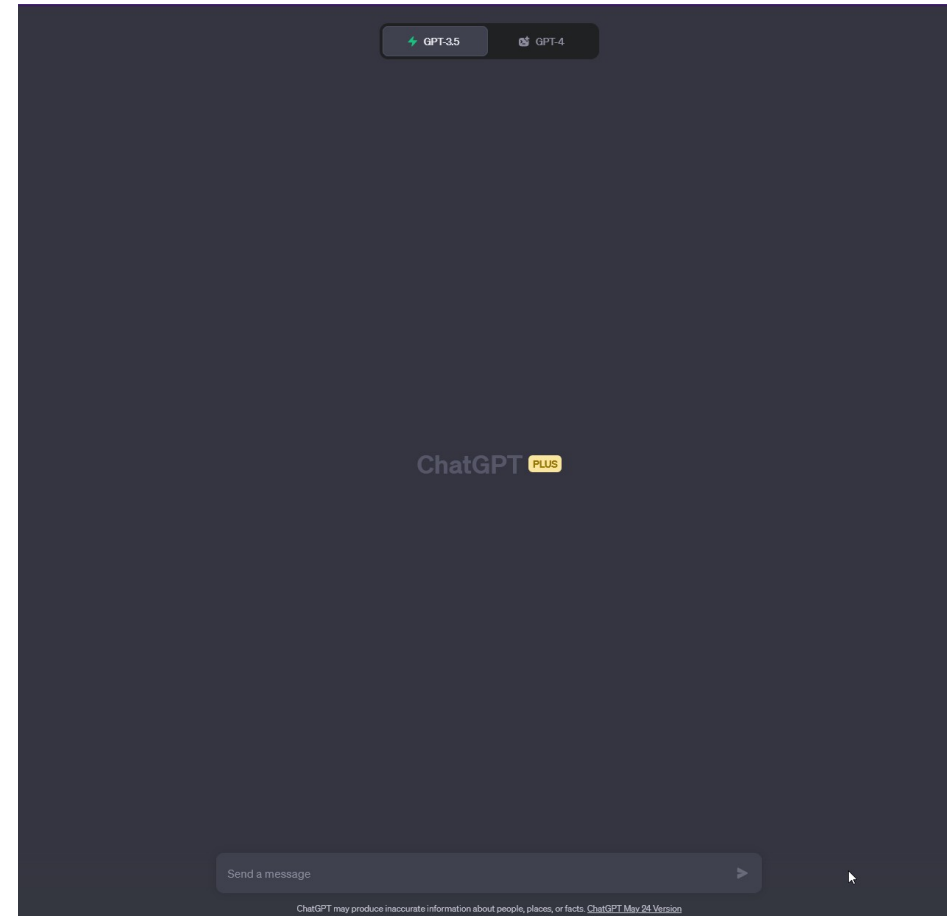


today we launched ChatGPT. try talking with it here:

chat.openai.com

7:38 PM · Nov 30, 2022

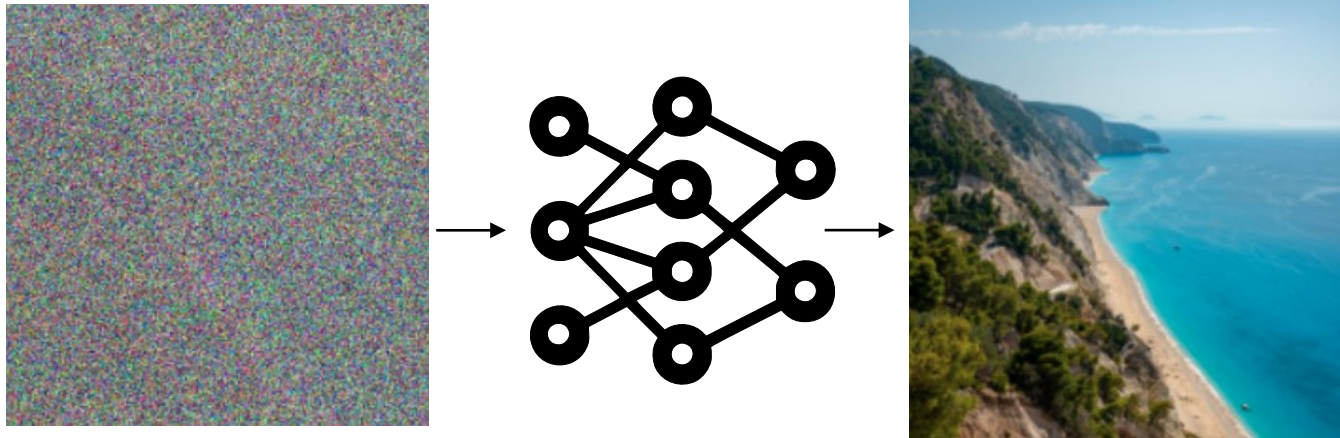
 1.2K  9.1K  34K  9.1K 



Generating images

- Modelling image distribution $p(y)$ is difficult

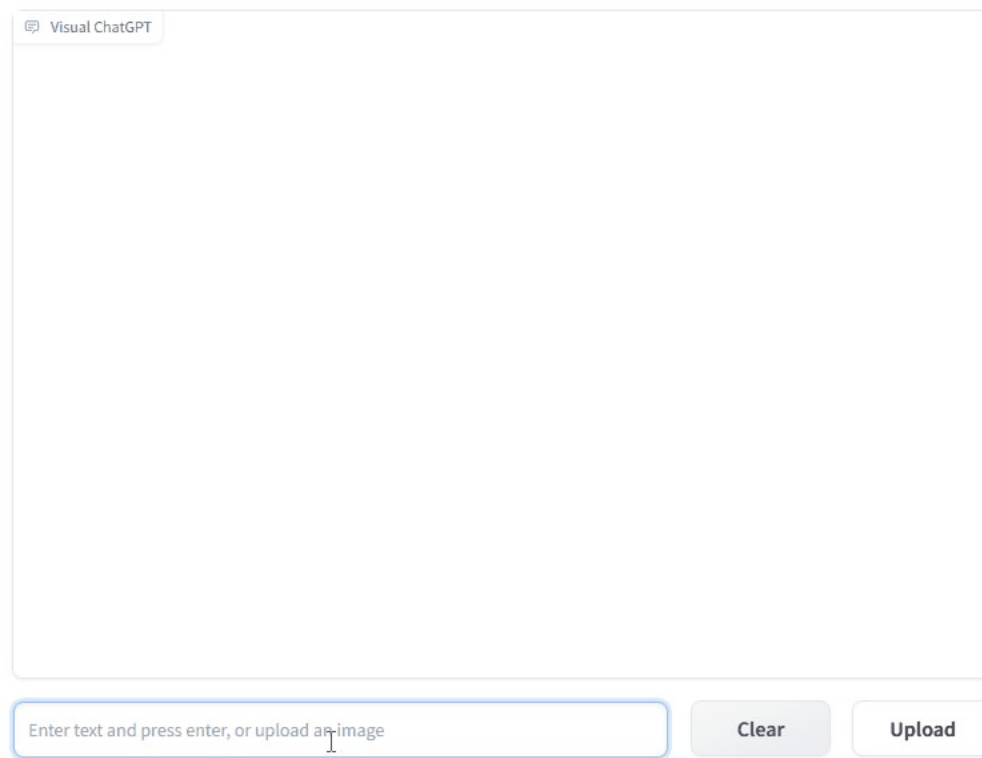
intractable



Diffusion Models (2022)



Generative Models (2022)



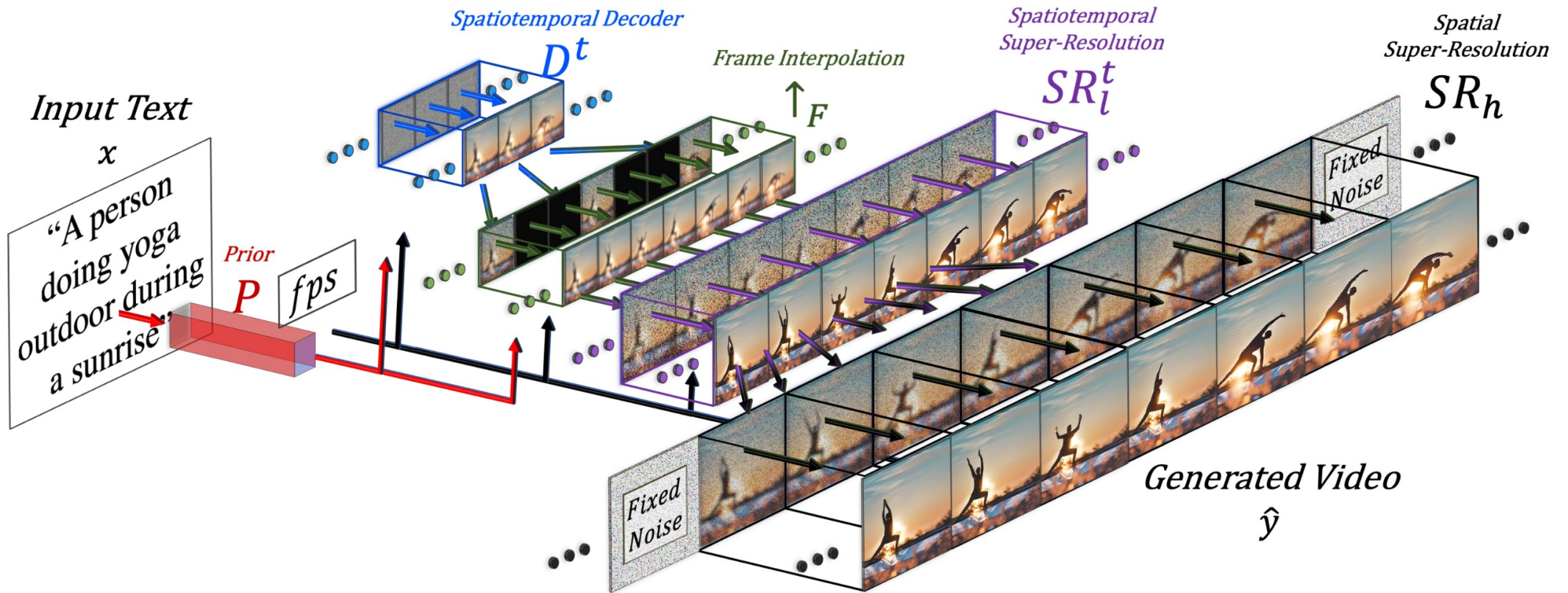
Visual ChatGPT

Enter text and press enter, or upload an image

Clear Upload

The image shows a web interface for 'Visual ChatGPT'. It features a large, empty rectangular area for chat messages. At the bottom, there is a text input field with the placeholder text 'Enter text and press enter, or upload an image'. To the right of the input field are two buttons: 'Clear' and 'Upload'.

Denoise all frames at once



memory expensive (limited to few seconds)

Generating Videos (2024)



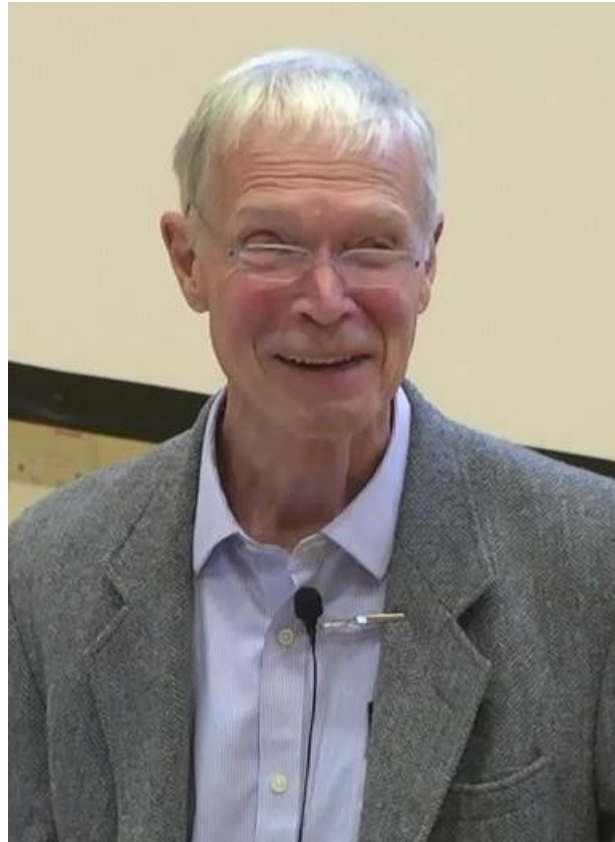
Nobel Prize in Physics (2024)

Hinton speaking at the Nobel Prize
Lectures in Stockholm in 2024



[This image](#) is [CC0 public domain](#)

John Hopfield in 2016



[This image](#) is [CC0 public domain](#)

In 2024, he was jointly awarded the [Nobel Prize in Physics](#) with [John Hopfield](#) “for foundational discoveries and inventions that enable machine learning with artificial neural networks.”

Nobel Prize in Chemistry (2024)

- The 2024 Nobel Prize in Chemistry was awarded to Demis Hassabis, co-founder of Google DeepMind, John Jumper, DeepMind Director, and David Baker for their revolutionary work in protein science.



David Baker

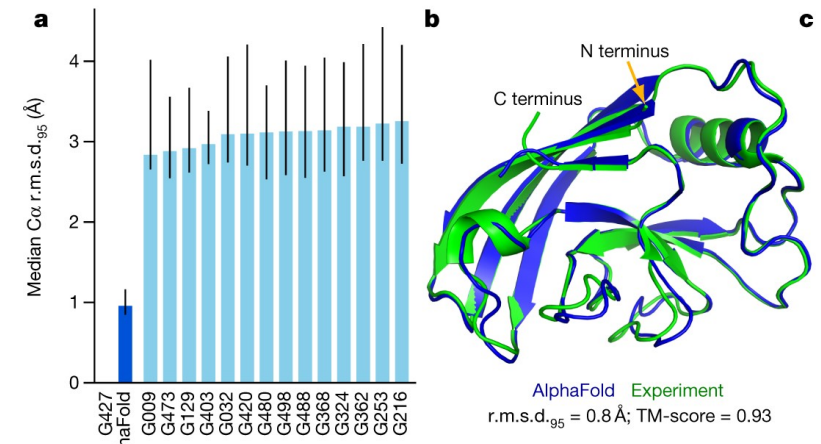


Demis Hassabis



John Jumper

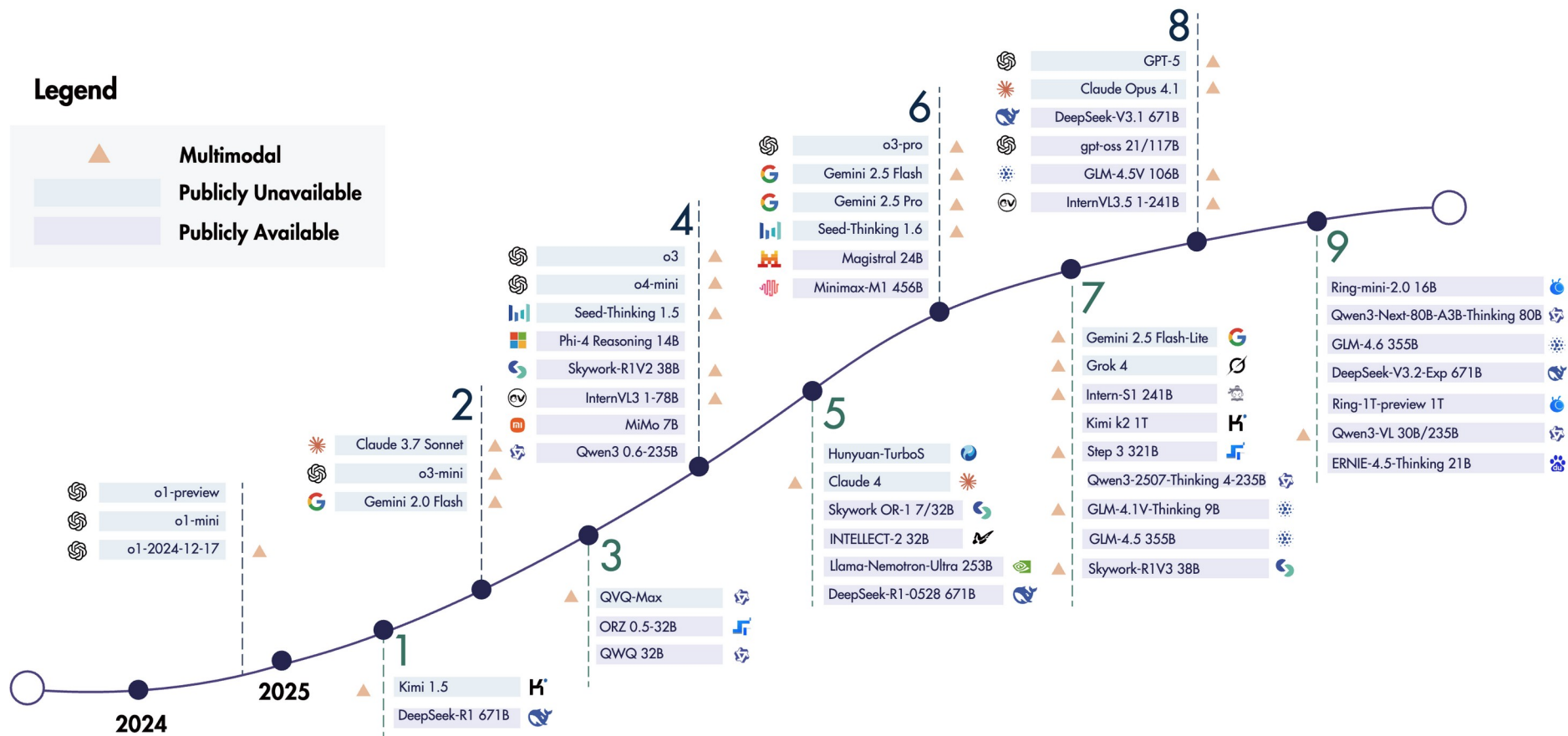
Photos: Arthur Petron, Nobel Prize week, Stockholm 2024, CC BY-SA 4.0



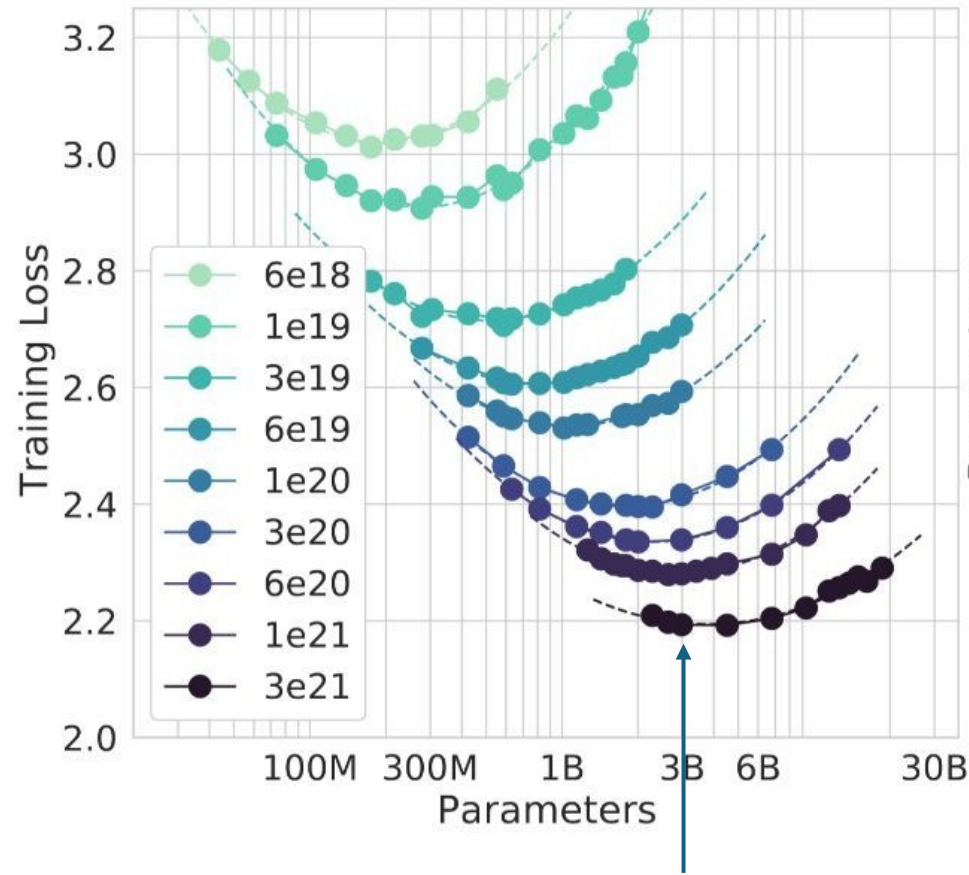
AlphaFold 2 vs the next-best CASP14 groups (left); prediction in blue, experiment in green (right)

Jumper et al., Nature 596, 2021, Fig. 1, CC BY 4.0

New LLMs every year (2023–2026)



Chinchilla law

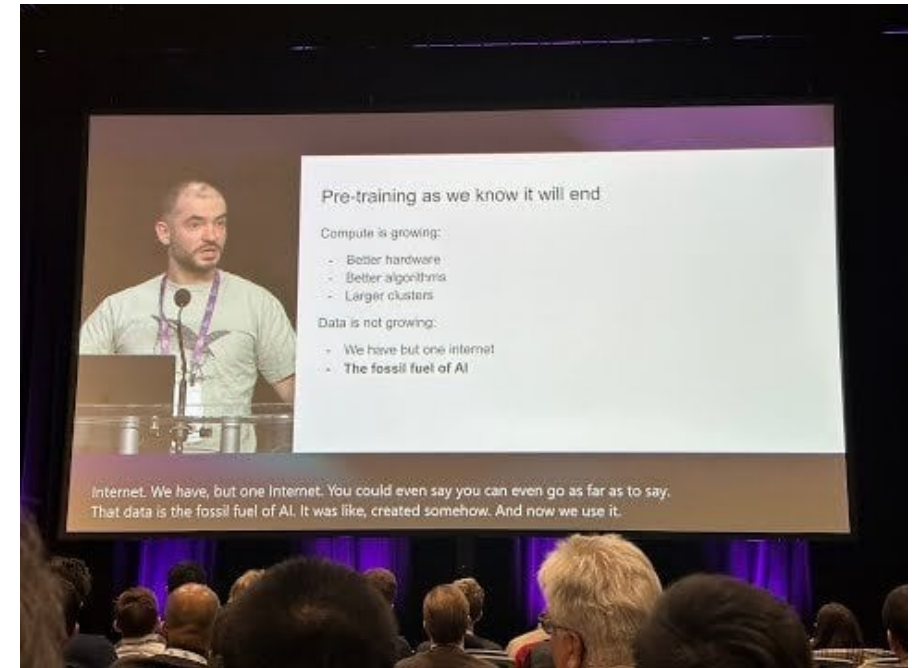


We observe the trend that more data leads to lower loss, but without increasing the data, there is a limit

Given #tokens, each model has an optimal number of parameters

Can pre-training continue?

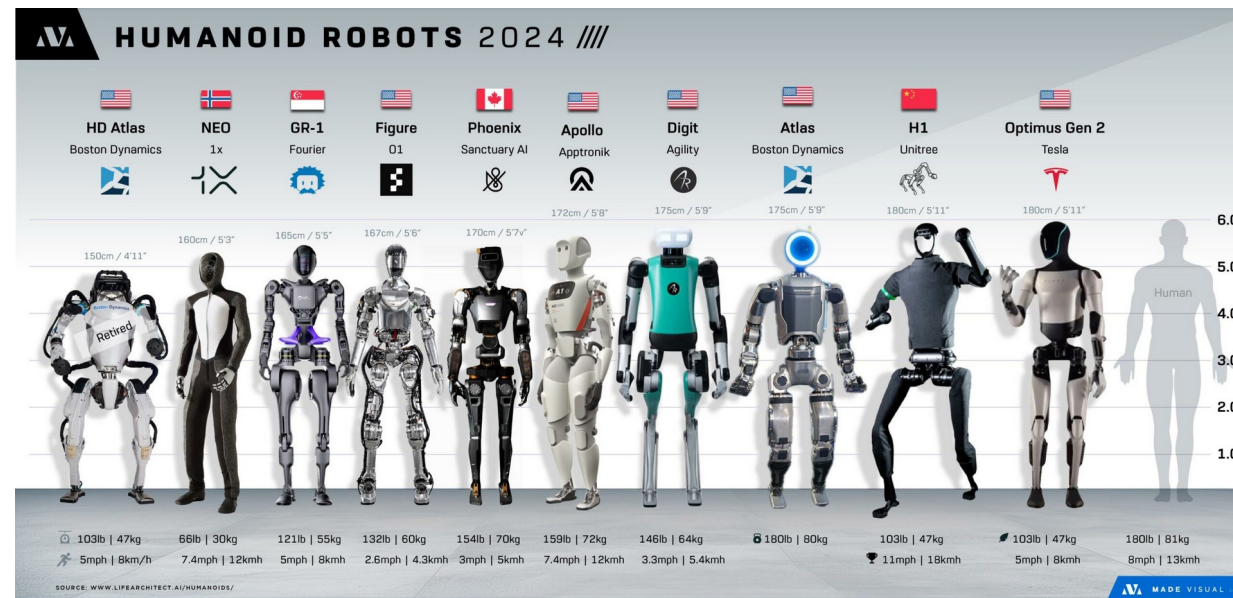
- Compute is growing:
 - Better hardware
 - Better algorithms
 - Larger clusters
- Data is not growing:
 - We have but one internet
 - The fossil fuel of AI
- **RL Post-training?**



Ilya Sutskever at Neurips'24

Humanoids (2027?)

- We want generalizing policies “What should I do next?” in the physical world^[1,2]



[1] - MINEDOJO; (Fan et al. NeurIPS'22)

[2] - ThinkAct: Vision-Language-Action Reasoning via Reinforced Visual Latent Planning; Chi-Pin (NeurIPS'25)

Video from <https://www.nvidia.com/en-eu/ai/cosmos/>

The Course

- Assignments are optional
- Hands-on tutorials
- Related Lectures:
 - DL for CV (cs231n)
 - Years 2014-2019
 - LLMs (cs336)
 - Years 2023-2026